

# Self similar diffusion

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## 1 The self-similar framework

### 1.1 Rescaling into a self-similar frame

Let  $u_i^*(\mathbf{x}^*, t^*)$ ,  $i = 1, \dots, d$ , satisfy an evolution equation of the form

$$\frac{\partial u_i^*}{\partial t^*} = R_i^*(u^*), \quad (1)$$

on  $\mathbb{R}^d$ . We rescale amplitude, length, and time by

$$u_i = \frac{u_i^*}{U}, \quad x_j = \frac{x_j^*}{L}, \quad t = \frac{t^*}{s(t)}, \quad (2)$$

where the amplitude scale  $U$  and the length scale  $L$  depend on the non dimensional time  $t$ . The timescale is determined by specifying the clock

$$\tau = \frac{dt^*}{dt}. \quad (3)$$

The clock  $\tau$  is problem specific and can contain the time dependent length and amplitude scales. It is defined in §4.1 and §5.1 for the two cases considered here.

Throughout, a prime denotes differentiation with respect to  $t$  and we use the following abbreviations

$$\sigma_L = \frac{L'}{L}, \quad \sigma_U = \frac{U'}{U}. \quad (4)$$

Differentiating  $x_j = x_j^*/L$  with respect to  $t$  at fixed  $x_j^*$  gives, for each component  $j$ ,

$$\frac{dx_j}{dt} = -x_j^* \frac{L'}{L^2} = -x_j \sigma_L. \quad (5)$$

Applying the chain rule to the left-hand side of (1), and using (3) and (5),

$$\begin{aligned} \frac{\partial u_i^*}{\partial t^*} &= \frac{dt}{dt^*} \left( \frac{\partial(Uu_i)}{\partial t} + U \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} \right) \\ &= \frac{U}{\tau} \left( \frac{\partial u_i}{\partial t} + \sigma_U u_i - \sigma_L x_j \frac{\partial u_i}{\partial x_j} \right). \end{aligned} \quad (6)$$

Equating (6) to  $R_i^*$  and solving for  $\partial u_i / \partial t$  gives the result used throughout this document. Under the rescaling (2), the equation (1) becomes

$$\frac{\partial u_i}{\partial t} = \underbrace{\sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i}_{\text{scaling terms}} + \underbrace{\frac{\tau}{U} R_i^*(Uu, Lx)}_{\text{physical terms}}. \quad (7)$$

Accordingly we split (7) into two flows. The *scaling operator*  $\Phi_{\Delta t}^{(S)}$  advances the scaling terms,

$$\Phi_{\Delta t}^{(S)}(u_i) = u_i + \int_t^{t+\Delta t} \left( \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i \right) dt, \quad (8)$$

and the *physics operator*  $\Phi_{\Delta t}^{(P)}$  advances the rest,

$$\Phi_{\Delta t}^{(P)}(u_i) = u_i + \int_t^{t+\Delta t} \frac{\tau}{U} R_i^*(Uu, Lx) dt. \quad (9)$$

Both are written as the field plus the integral of the terms they advance.  $\Phi^{(S)}$  is obtained in closed form below and is shared;  $\Phi^{(P)}$  is equation-specific and is supplied in §4.2 and §5.6.

## 1.2 The scaling operator

Evaluating  $\Phi_{\Delta t}^{(S)}$  amounts to solving

$$\frac{\partial u_i}{\partial t} = \sigma_L x_j \frac{\partial u_i}{\partial x_j} - \sigma_U u_i. \quad (10)$$

This is a linear, first-order PDE and can be solved by the method of characteristics. Note that the scaling terms do not couple the components of  $u_i$ , so the argument below is componentwise.

Let  $x_j(t)$  denote a characteristic curve satisfying

$$\frac{dx_j}{dt} = -x_j \sigma_L. \quad (11)$$

The total derivative of  $u_i$  along such a curve is

$$\frac{du_i}{dt} = \frac{\partial u_i}{\partial t} + \frac{\partial u_i}{\partial x_j} \frac{dx_j}{dt} = \frac{\partial u_i}{\partial t} - \sigma_L x_j \frac{\partial u_i}{\partial x_j}, \quad (12)$$

so along the characteristic (10) reduces to the ordinary differential equation

$$\frac{du_i}{dt} = -\sigma_U u_i. \quad (13)$$

Integrating (11) from  $t_k$  to  $t_{k+1}$  gives

$$\ln x_j(t_{k+1}) - \ln x_j(t_k) = -[\ln L_{k+1} - \ln L_k] \implies x_j(t_{k+1}) = x_j(t_k) \frac{L_k}{L_{k+1}}, \quad (14)$$

so that the characteristic arriving at the evaluation point  $x_j \equiv x_j(t_{k+1})$  originated at

$$x_j(t_k) = x_j \frac{L_{k+1}}{L_k}. \quad (15)$$

Integrating (13) gives

$$u_i(t_{k+1}) = u_i(t_k) U_k / U_{k+1}. \quad (16)$$

Combining this with the characteristic origin (15) and the initial condition  $u_i(t_k, \cdot)$  gives the scaling operator in closed form,

$$\Phi_{\Delta t}^{(S)}(u_i)(\mathbf{x}) = \frac{U_k}{U_{k+1}} u_i\left(\mathbf{x} \frac{L_{k+1}}{L_k}\right). \quad (17)$$

Writing the step ratios

$$\lambda_{U,k+1} = \frac{U_{k+1}}{U_k}, \quad \lambda_{L,k+1} = \frac{L_{k+1}}{L_k}, \quad (18)$$

this is

$$\Phi_{\Delta t}^{(S)}(u_i)(\mathbf{x}) = \frac{1}{\lambda_{U,k+1}} u_i(\mathbf{x} \lambda_{L,k+1}). \quad (19)$$

### 1.2.1 The scaling operator in $\kappa$ -space

For equations formulated spectrally it is useful to have the transform of the scaling terms. Using  $\widehat{\cdot}$  for the spatial Fourier transform,

$$\hat{u}_i(\boldsymbol{\kappa}) = \int u_i(\mathbf{x}) e^{-i\boldsymbol{\kappa}_j x_j} d\mathbf{x}, \quad (20)$$

and writing the spread term in conservative form,

$$x_j \frac{\partial u_i}{\partial x_j} = \frac{\partial}{\partial x_j} (x_j u_i) - d u_i, \quad (21)$$

gives

$$\widehat{x_j \frac{\partial u_i}{\partial x_j}} = i\kappa_j \left( i \frac{\partial \hat{u}_i}{\partial \kappa_j} \right) - d \hat{u}_i = - \left( \kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d \hat{u}_i \right), \quad (22)$$

so that the transform of (10) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left( \kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d \hat{u}_i \right) - \sigma_U \hat{u}_i. \quad (23)$$

This is again a linear, first-order PDE, now with characteristics in  $\boldsymbol{\kappa}$  satisfying

$$\frac{d\kappa_j}{dt} = \kappa_j \sigma_L, \quad (24)$$

along which (23) reduces to

$$\frac{d\hat{u}_i}{dt} = -(d\sigma_L + \sigma_U) \hat{u}_i. \quad (25)$$

Integrating (24) from  $t_k$  to  $t_{k+1}$  gives  $\kappa_j(t_{k+1}) = \kappa_j(t_k) \lambda_{L,k+1}$ , so the characteristic arriving at  $\boldsymbol{\kappa}$  originated at  $\boldsymbol{\kappa}/\lambda_{L,k+1}$ , while integrating (25) gives  $\hat{u}_i(t_{k+1}) = \hat{u}_i(t_k)/(\lambda_{U,k+1} \lambda_{L,k+1}^d)$ . The scaling operator in  $\boldsymbol{\kappa}$ -space is therefore

$$\widehat{\Phi_{\Delta t}^{(S)}(u_i)}(\boldsymbol{\kappa}) = \frac{1}{\lambda_{U,k+1} \lambda_{L,k+1}^d} \hat{u}_i \left( \frac{\boldsymbol{\kappa}}{\lambda_{L,k+1}} \right). \quad (26)$$

This agrees with transforming (19) directly: the substitution  $\mathbf{x}' = \mathbf{x} \lambda_{L,k+1}$  in (20) contributes the Jacobian  $\lambda_{L,k+1}^{-d}$  and maps  $\boldsymbol{\kappa}$  to  $\boldsymbol{\kappa}/\lambda_{L,k+1}$ . A growing length scale ( $\lambda_{L,k+1} > 1$ ) thus compresses  $u_i$  toward  $\mathbf{x} = 0$  and correspondingly stretches  $\hat{u}_i$  toward larger  $|\boldsymbol{\kappa}|$ .

## 1.3 Spectral interpolation

The closed form (19) requires  $u_i$  at the rescaled points  $\mathbf{x} \lambda_{L,k+1}$ , which are not grid points. On a uniform periodic grid the represented field is band-limited, so it can be evaluated at arbitrary points exactly, with no interpolation error, by summing its truncated Fourier series there:

$$u_i(\mathbf{x} \lambda_{L,k+1}) = \sum_{\boldsymbol{\kappa} \in \mathcal{K}} \hat{u}_i(\boldsymbol{\kappa}) \exp(i \boldsymbol{\kappa}_j x_j \lambda_{L,k+1}), \quad (27)$$

where  $\mathcal{K}$  is the set of represented wavenumbers and  $\hat{u}_i(\boldsymbol{\kappa})$  are the normalized transform coefficients of  $u_i$ . This is the defining property of the scheme: the scaling step introduces no spatial discretization error of its own.

**Queries outside the box.** The true problem we are trying to approximate is posed on an infinite box, with  $u_i$  going to zero as it approaches the boundaries. For  $\lambda_{L,k+1} > 1$  it is possible that  $|x_j \lambda_{L,k+1}| > H/2$ , in which case we would like the rescaling in (27) to return 0. However, this is not the current behavior: instead, it wraps around the periodic box and returns a potentially nonzero value from the other end of the domain. Therefore, to obtain the desired behavior we modify the scaling step to

$$u_i(\mathbf{x} \lambda_{L,k+1}) = \chi(\mathbf{x} \lambda_{L,k+1}) \sum_{\boldsymbol{\kappa} \in \mathcal{K}} \hat{u}_i(\boldsymbol{\kappa}) \exp(i \boldsymbol{\kappa}_j x_j \lambda_{L,k+1}), \quad \chi(\mathbf{y}) = \prod_d \mathbb{I}[|y_d| \leq H/2]. \quad (28)$$

## 1.4 Constraint functionals

We will set the ratios  $\lambda_{U,k+1}, \lambda_{L,k+1}$  that the scaling step (19) depends on by requiring the rescaled field to carry prescribed values of two functionals  $F_1, F_2$ . Each is assumed to be a finite sum of terms that are individually homogeneous under the scaling operator,

$$F_m \left[ \frac{1}{\lambda_U} u(x\lambda_L) \right] = \sum_j \lambda_U^{-p_{mj}} \lambda_L^{-q_{mj}} F_{mj}[u], \quad m = 1, 2, \quad (29)$$

with  $F_m = \sum_j F_{mj}$  and each  $F_{mj}[u] > 0$ . The targets  $F_1, F_2$  are fixed once, from the initial condition (§2.2). Given a field  $u$ , its per-term values  $F_{mj}^{\text{in}} = F_{mj}[u]$  are evaluated once, after which the right-hand side of (29) is an explicit function of  $a = \ln \lambda_U$  and  $b = \ln \lambda_L$  alone: no further evaluation of the scaled field is needed to solve for the ratios. Imposing the targets in logarithmic form gives the residuals

$$R_m(a, b) = \ln \sum_j F_{mj}^{\text{in}} e^{-p_{mj}a - q_{mj}b} - \ln F_m = 0, \quad m = 1, 2. \quad (30)$$

Writing  $w_{mj}(a, b)$  for the fraction of  $F_m$  carried by term  $j$  at the current ratios, the Jacobian of (30) is minus the matrix of weighted exponents,

$$\frac{\partial(R_1, R_2)}{\partial(a, b)} = - \begin{bmatrix} \bar{p}_1 & \bar{q}_1 \\ \bar{p}_2 & \bar{q}_2 \end{bmatrix}, \quad \bar{p}_m = \sum_j w_{mj} p_{mj}, \quad \bar{q}_m = \sum_j w_{mj} q_{mj}, \quad w_{mj} = \frac{F_{mj}^{\text{in}} e^{-p_{mj}a - q_{mj}b}}{\sum_l F_{ml}^{\text{in}} e^{-p_{ml}a - q_{ml}b}}, \quad (31)$$

and (30) is solved by Newton's method on  $(a, b)$ . The system is uniquely solvable near the root as long as  $\bar{p}_1 \bar{q}_2 - \bar{p}_2 \bar{q}_1 \neq 0$ ; each functional must respond to amplitude and to length rescaling differently from the other. Since  $\lambda_U$  and  $\lambda_L$  stay close to unity, Newton started from  $a = b = 0$  converges in a few iterations; we iterate until  $\max_m |R_m| < 10^{-12}$  and treat a failure to do so within 20 iterations as an error.

When each functional consists of a single term, the residuals (30) are linear in  $(a, b)$  and Newton converges in one step. The constraints then reduce to the linear system

$$\begin{bmatrix} p_1 & q_1 \\ p_2 & q_2 \end{bmatrix} \begin{bmatrix} \ln \lambda_U \\ \ln \lambda_L \end{bmatrix} = \begin{bmatrix} \ln F_1^{\text{in}} - \ln F_1 \\ \ln F_2^{\text{in}} - \ln F_2 \end{bmatrix}, \quad (32)$$

uniquely solvable when  $p_1 q_2 - p_2 q_1 \neq 0$ , which is solved in closed form instead.

In both cases the second functional is a second moment, and we normalize it by the target value of the zeroth moment  $F_1$ . As the normalization is a constant, it changes neither the exponents nor the ratios the constraints produce; it gives  $F_2$  the units of a squared length, a mean-square width. The functionals themselves are equation-specific (§4.3, §5.8); the substep that applies (30) is `CONSTRAINEDSCALE` (§2.5).

## 1.5 Power-law ansatz and the step in physical time

The algorithm also needs to *predict* scale ratios over a fraction of a step, at points where no constraint is available to fix them (§2.7). For this we assume that both scales follow a power law in physical time,

$$X = (t^* + \alpha_X)^{\beta_X}, \quad X \in \{U, L\}, \quad (33)$$

so that the ratio over a fraction  $i$  of step  $k$  is

$$\frac{X_{k+i}}{X_k} = \left( 1 + \frac{i \Delta t_k^*}{t_k^* + \alpha_X} \right)^{\beta_X}. \quad (34)$$

Since the time step  $\Delta t_k = t_{k+1} - t_k$  is taken in the self-similar time  $t$ , the corresponding physical increment must be obtained from the clock,

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau \, dt, \quad (35)$$

which can be approximated with a three-point Adams–Bashforth-style quadrature:

$$\Delta t_k^* = \int_{t_k}^{t_{k+1}} \tau \, dt \approx w_0 \tau_k + w_1 \tau_{k-1} + w_2 \tau_{k-2}. \quad (36)$$

The coefficients can be computed as follows.

$$\begin{aligned} w_0 &= \frac{h^3/3 + (2h_1 + h_2)h^2/2 + h_1(h_1 + h_2)h}{h_1(h_1 + h_2)}, \\ w_1 &= -\frac{h^3/3 + (h_1 + h_2)h^2/2}{h_1 h_2}, \\ w_2 &= \frac{h^3/3 + h_1 h^2/2}{h_2(h_1 + h_2)}, \end{aligned} \quad (37)$$

where  $h = \Delta t_k$ ,  $h_1 = \Delta t_{k-1}$ ,  $h_2 = \Delta t_{k-2}$ .

Eliminating  $t_k^* + \alpha_X = X_k^{1/\beta_X}$  from (33) and taking logarithms gives the function that the algorithm uses in both directions,

$$\ln \frac{X_{k+i}}{X_k} = G_X(i, \beta_X; k), \quad G_X(i, \beta; k) = \beta \ln \left( 1 + \exp \left( \ln i + \ln \Delta t_k^* - \frac{\ln X_k}{\beta} \right) \right). \quad (38)$$

It is evaluated forward, with  $i = \frac{1}{2}$ , to predict a half-step ratio (§2.5), and inverted for  $\beta$ , with  $i = 1$ , to estimate the exponents from the realized history (§2.8).

**Logarithmic arithmetic.**  $U_k$  and  $L_k$  may become extremely large or small, so we store  $\ln U_k$  and  $\ln L_k$  rather than  $U_k$  and  $L_k$ , and perform all arithmetic on them in the logarithmic number system, where multiplication and division become addition and subtraction, and where sums and differences are evaluated by

$$\log_b(X + Y) = \log_b(X) + \log_b(1 + b^{y-x}), \quad \log_b(X - Y) = \log_b(X) + \log_b(1 - b^{y-x}) \quad (X > Y > 0), \quad (39)$$

with  $x = \log_b(X)$ ,  $y = \log_b(Y)$ . In particular (36) is evaluated as

$$\ln \Delta t_k^* = \ln(w_0 e^{\ln \tau_k} + w_1 e^{\ln \tau_{k-1}} + w_2 e^{\ln \tau_{k-2}}). \quad (40)$$

## 2 Algorithm

The integrator below is stated once, for any equation that supplies the three items of §1: a clock  $\tau$ , a physics operator  $\Phi^{(P)}$ , and a pair of constraint functionals. Everything else — the splitting, the scale bookkeeping, the exponent estimation — is common.

### 2.1 Domain, grid, and notation

The spatial domain is periodic with a fixed physical side length  $H$  and  $N$  grid points per direction, centred at the origin, so that  $x_j \in [-H/2, H/2)$  with uniform spacing  $\Delta x = H/N$  and grid points

$$x_n = -\frac{H}{2} + n \Delta x, \quad n = 0, 1, \dots, N-1 \quad (41)$$

in each of the  $d$  directions. Every integral over the domain below is approximated by the Riemann sum

$$\int f(\mathbf{x}) \, d\mathbf{x} \approx (\Delta x)^d \sum_{\mathbf{n}} f(\mathbf{x}_{\mathbf{n}}) \quad (42)$$

on this grid.

An index range abbreviates the corresponding tuple of stored values, e.g.  $\ln \tau_{k-2:k} = (\ln \tau_{k-2}, \ln \tau_{k-1}, \ln \tau_k)$  and  $\Delta t_{k-2:k} = (\Delta t_{k-2}, \Delta t_{k-1}, \Delta t_k)$ . The run constants  $H, N$ , the time-step constants  $\Delta t_{\max}, C, \gamma, \Delta t_{\min}$  of §2.3, the constraint targets  $F_1, F_2$ , and whatever physical parameters the equation carries are visible to every subroutine and are omitted from their signatures. The step  $\Delta t_k$  is updated dynamically. As set out in §1.5, all arithmetic on the scales is done on  $\ln U_k, \ln L_k$ .

## 2.2 Initialization

A run is specified by the initial field  $u_0$ , the domain parameters  $H, N$ , the time-step constants  $\Delta t_{\max}, C, \gamma, \Delta t_{\min}$  of §2.3, a time horizon  $T_{\text{end}}$ , the splitting scheme  $scheme \in \{\text{LIETROTTER}, \text{STRANG}\}$  (§2.6, §2.7), for equations that define one, the switch *project* of the optional projection inside SCALE (§2.5), and initial guesses  $\beta_U^{(0)}, \beta_L^{(0)}$ , which only seed the bracket of the first root-find for  $\beta_U, \beta_L$  (§2.8). From these:

- The integration loop starts at  $t_0 = 0$  and runs while  $t_k < T_{\text{end}}$ .
- The self-similar frame is initialized at the initial condition itself,  $L_0 = U_0 = 1$ , i.e.  $\ln L_0 = \ln U_0 = 0$ .
- The *target* values  $F_1, F_2$  of the constraint functionals (§1.4) are fixed once, from  $u_0$  itself,

$$F_1 = F_1[u_0], \quad F_2 = F_2[u_0]. \quad (43)$$

- The flag  $\beta_{\text{ok}}$ , which records whether the current  $\beta_U, \beta_L$  are usable by STRANG, starts as  $\beta_{\text{ok}} = \text{false}$ .

## 2.3 Time step

The step  $\Delta t_k$  is chosen at the start of each step from the field  $u_k$  it starts from,

$$\Delta t_k = \min(\Delta t_{\max}, \gamma \Delta t_{k-1}, \Delta t_{\text{CFL}}(u_k; C)), \quad (44)$$

where  $\Delta t_{\text{CFL}}(u; C) = \text{MAXSTABLEDT}(u, C)$  (§2.5) is the largest step at which the equation's physics operator runs at the target CFL number  $C$ , expressed as a fraction of the time integrator's own stability limit. For Navier–Stokes this is the advective bound (170). The three terms play distinct roles.

- $\Delta t_{\max}$  caps the step when the field is quiet, and is the step of the settled regime.
- $\Delta t_{\text{CFL}}$  shrinks the step when the field concentrates: a momentary high-amplitude event raises  $\max |\mathbf{u}|$  and would otherwise carry the physics step past its stability limit.
- $\gamma > 1$  caps *growth only*. The step drops in a single step when an event arrives and recovers geometrically afterwards. Step  $k = 0$  has no predecessor and is uncapped. The cap keeps the step ratios  $h/h_1, h_1/h_2$  in the quadrature (37) moderate, and it limits the per-step growth of  $L$ , which drives the feedback between the predicted half step and the  $\beta$  fit under STRANG (§2.7). We take  $C = 0.5, \gamma = 1.1$ .

If (44) falls below  $\Delta t_{\min}$  the run stops with an error: a runaway  $\max |\mathbf{u}|$  should end the run rather than stall it in ever smaller steps. Fixed stepping at  $\Delta t_{\max}$  is the special case with no CFL term,  $\Delta t_{\text{CFL}} = \infty$ . It is what the 1D diffusion equation uses, since its propagator (68) is exact at any step and has no CFL limit to adapt to.

**No rejection.** Every step is accepted. The CFL is measured on  $u_k$ , so the step overshoots its target only if  $\max |\mathbf{u}|$  grows within the step; with  $C = 0.5$  it has to more than double in a single step to breach the stability limit. Whether that happens is monitored rather than prevented. The *realized* CFL of step  $k$ ,

$$\text{CFL}_k = \frac{\Delta t_k}{\Delta t_{\text{CFL}}(u_{k+1}; 1)}, \quad (45)$$

measures the step actually taken against the field it produced. It is recorded every step, and the run warns the first time it exceeds 1. The pre-step value would carry no information whenever the CFL term is active, since it reads back  $C$  by construction.

**The end of the run.** The loop runs while  $t_k < T_{\text{end}}$ , with  $t_{k+1} = t_k + \Delta t_k$ , and the last step is *not* shortened to land on  $T_{\text{end}}$ . A clipped step can be arbitrarily small, which would put an extreme ratio  $h/h_1$  into (37) on the next  $\beta$  update. The run overshoots  $T_{\text{end}}$  by less than one step, which is immaterial on the logarithmic clock  $t$ . For the same reason, output taken at prescribed times  $t$  is taken at the first step reaching each time, never by shortening a step.

## 2.4 Top-level integrator

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**Algorithm 1** Top-level integrator

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**Require:**  $u_0, H, N, \Delta t_{\max}, C, \gamma, \Delta t_{\min}, T_{\text{end}}, \text{scheme}, \beta_U^{(0)}, \beta_L^{(0)}$

- 1:  $t_0 \leftarrow 0, k \leftarrow 0$
- 2:  $\ln U_0 \leftarrow 0, \ln L_0 \leftarrow 0$
- 3: Compute constraint targets  $F_1, F_2$  from  $u_0$  via (43)
- 4:  $\beta_U \leftarrow \beta_U^{(0)}, \beta_L \leftarrow \beta_L^{(0)}, \beta_{\text{ok}} \leftarrow \text{false}$
- 5: Store  $(u_0, \ln U_0, \ln L_0, \ln \tau_0, \beta_U, \beta_L, \beta_{\text{ok}})$
- 6: **while**  $t_k < T_{\text{end}}$  **do**
- 7:    $\Delta t_k \leftarrow (44)$  on  $u_k$ ; **error** if  $\Delta t_k < \Delta t_{\min}$  ▷ §2.3
- 8:   **if**  $\text{scheme} = \text{STRANG}$  **and**  $\beta_{\text{ok}}$  **then**
- 9:      $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{STRANG}(u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L)$  ▷ §2.7
- 10:   **else**
- 11:      $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}) \leftarrow \text{LIETROTTER}(u_k, \ln U_k, \ln L_k, \Delta t_k)$  ▷ §2.6
- 12:   **end if**
- 13:    $u_{k+1} \leftarrow \text{APPLYBC}(u_{k+1})$  ▷ equation-specific, §2.5
- 14:    $\ln \tau_{k+1} \leftarrow \ln \tau(\ln U_{k+1}, \ln L_{k+1})$  ▷ (65), (86)
- 15:    $t_{k+1} \leftarrow t_k + \Delta t_k$
- 16:   **if**  $k + 1 \geq 3$  **then**
- 17:      $(\beta_U, \beta_L, \beta_{\text{ok}}) \leftarrow \text{ESTIMATEBETA}(\ln U_{k:k+1}, \ln L_{k:k+1}, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L)$  ▷ §2.8
- 18:   **end if**
- 19:   Store  $(u_{k+1}, \ln U_{k+1}, \ln L_{k+1}, \ln \tau_{k+1}, t_{k+1}, \Delta t_k, \text{CFL}_k, \beta_U, \beta_L, \beta_{\text{ok}})$  ▷ (45)
- 20:    $k \leftarrow k + 1$
- 21: **end while**

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The run parameter *scheme* selects the step function. STRANG needs usable estimates of  $\beta_U, \beta_L$ , so the loop falls back to LIETROTTER whenever  $\beta_{\text{ok}}$  is false. This is always the case for the first three steps  $k = 0, 1, 2$ : the first  $\beta$  update needs four stored states, so it first runs at the end of step  $k = 2$ . It is also the case for every step that follows a failed  $\beta$  update (§2.8), until a later update succeeds. The  $\beta$  update itself runs regardless of *scheme*; under *scheme* = LIETROTTER the estimates are diagnostic only.

## 2.5 Substeps

Both splitting schemes are assembled from the substeps below. None of them applies the boundary treatment, which is handled only by the top-level loop.

**Evolve**( $u, \Delta t_k$ ). Applies the physics operator  $\Phi_{\Delta t_k}^{(\text{P})}$  of (9): it advances the non-scaling terms of (7) over  $\Delta t_k$ , leaving the scales  $\ln U, \ln L$  untouched.

**MaxStableDt**( $u, C$ ). Equation-specific: the largest step at which EVOLVE from  $u$  runs at the fraction  $C$  of its stability limit, the term  $\Delta t_{\text{CFL}}(u; C)$  of (44). For Navier–Stokes it is (170). An equation without a CFL limit does not define it and runs at fixed  $\Delta t_{\max}$  (§2.3).

**Scale**( $u, \ln \lambda_U, \ln \lambda_L$ ). Applies the scaling operator (19) with prescribed ratios  $\lambda_U, \lambda_L$ , returning  $\lambda_U^{-1} u(\mathbf{x}_n \lambda_L)$  at each grid point, evaluated with the spectral interpolation of §1.3 and set to zero wherever  $\mathbf{x}_n \lambda_L$  leaves the box (28). Here, and wherever the substeps are used,  $\lambda_U$  and  $\lambda_L$  denote the ratio of  $U$ , respectively  $L$ , after the substep to its value before it. Inside a Lie–Trotter step this is  $\lambda_{U,k+1} = U_{k+1}/U_k$  of (18); inside a Strang step it is a ratio across a half step,  $U_{k+1/2}/U_k$  or  $U_{k+1}/U_{k+1/2}$  (§2.7).

**Optional projection.** With the run parameter *project* enabled, SCALE ends by applying an equation-specific projection  $\mathcal{P}$  onto the admissible fields, returning  $\mathcal{P}(\lambda_U^{-1} u(\mathbf{x}_n \lambda_L))$ ; with it disabled, or for an equation that defines no  $\mathcal{P}$ , the projection is the identity. The continuous operator (19) maps a divergence-free field

to a divergence-free field, but its evaluation on the grid by interpolation and the out-of-box cutoff does not preserve this exactly. For Navier–Stokes,  $\mathcal{P}$  is the Leray projection (146), which restores  $\nabla \cdot \tilde{\mathbf{u}} = 0$  (§5.10); the 1D diffusion equation defines none. Because every scaling step runs through SCALE, the projection follows the scaling step of LIETROTTER and *both* scaling half steps of STRANG. Inside CONSTRAINEDSCALE it is applied after the ratios have been solved, so the returned field meets the targets  $F_1, F_2$  only up to the part the projection removes. The miss does not accumulate: the next CONSTRAINEDSCALE measures the functionals of the projected field afresh and absorbs it into the next ratios.

**ConstrainedScale**( $u, \ln U, \ln L$ ). Rescales  $u$  so that the result carries the fixed targets  $F_1, F_2$  of (43), and advances the log-scale state accordingly:

1. **Functionals of the input**, the per-term values  $F_{mj}^{\text{in}} = F_{mj}[u]$  of (29), evaluated on the grid via (42).
2. **Ratios**  $\ln \lambda_U, \ln \lambda_L$  from the residuals (30), by Newton’s method, or in closed form from (32) when each functional is a single term. This works on the scalars  $F_{mj}^{\text{in}}$  alone and does not touch the field.
3. **Rescale**,  $u' = \text{SCALE}(u, \ln \lambda_U, \ln \lambda_L)$ .
4. **Return**  $u'$  together with the advanced state  $\ln U + \ln \lambda_U, \ln L + \ln \lambda_L$ .

No time increment enters CONSTRAINEDSCALE: the ratios are fixed by the functionals of the field it is given, not by the elapsed time. The same substep therefore serves as the full scaling step of Lie–Trotter and as the closing scaling half step of Strang.

**PredictScale**( $i, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L$ ). Predicts the ratios over the fraction  $i$  of step  $k$  from the power-law form (33) rather than from the constraints. It evaluates  $\ln \Delta t_k^*$  from (40), with the weights (37) of the steps  $\Delta t_{k-2:k}$  — the last of which is the step about to be taken — and returns  $(G_U(i, \beta_U; k), G_L(i, \beta_L; k))$  using (38). The same function  $G_X$  is inverted for  $\beta$  in §2.8.

**ApplyBC**( $u$ ). Imposes whatever boundary treatment the equation requires, after the full step has been assembled. Both equations treated here clamp the domain boundary to zero: the 1D diffusion equation its two edge grid points (§4.5), the Navier–Stokes equations the six faces of the box (§5.11). APPLYBC performs no projection; that is done only inside SCALE. For Navier–Stokes the clamp is itself optional: with it disabled APPLYBC is the identity, and the far-field condition is left to the sponge forcing of §5.3, which acts inside EVOLVE.

## 2.6 LieTrotter: Lie–Trotter splitting

A Lie–Trotter step composes a full physics step with a full scaling step,  $u_{k+1} = \Phi_{\Delta t_k}^{(S)}(\Phi_{\Delta t_k}^{(P)}(u_k))$ , with the scaling ratios fixed by the constraints. It uses no estimate of  $\beta_U, \beta_L$ , so it is valid from the first step. It is written for a general input state  $(u, \ln U, \ln L)$  and step  $\Delta t_k$  rather than only a stored state  $(u_k, \ln U_k, \ln L_k)$ , so that STRANG can call it on a half-stepped state.

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### Algorithm 2 Lie–Trotter step

---

- 1: **function** LIETROTTER( $u, \ln U, \ln L, \Delta t_k$ )
  - 2:      $u^{(P)} \leftarrow \text{EVOLVE}(u, \Delta t_k)$  ▷ §2.5
  - 3:     **return** CONSTRAINEDSCALE( $u^{(P)}, \ln U, \ln L$ ) ▷ §2.5
  - 4: **end function**
- 

## 2.7 Strang: Strang splitting

A Strang step is the symmetric composition

$$u_{k+1} = \Phi_{\Delta t_k/2}^{(S)}\left(\Phi_{\Delta t_k}^{(P)}\left(\Phi_{\Delta t_k/2}^{(S)}(u_k)\right)\right) \quad (46)$$

of a scaling half step, a full physics step, and a second scaling half step. The last two of these are exactly a Lie–Trotter step — a full physics step followed by a scaling step whose ratios come from the constraints — so STRANG is a predicted scaling half step followed by a call to LIETROTTER.

---

**Algorithm 3** Strang step

---

```

1: function STRANG( $u_k, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L$ )
2:    $(\ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k)) \leftarrow \text{PREDICTSCALE}(\frac{1}{2}, \ln U_k, \ln L_k, \ln \tau_{k-2:k}, \Delta t_{k-2:k}, \beta_U, \beta_L)$   $\triangleright$  (47)
3:    $u_{k+1/2} \leftarrow \text{SCALE}(u_k, \ln(U_{k+1/2}/U_k), \ln(L_{k+1/2}/L_k))$ 
4:    $\ln U_{k+1/2} \leftarrow \ln U_k + \ln(U_{k+1/2}/U_k); \ln L_{k+1/2} \leftarrow \ln L_k + \ln(L_{k+1/2}/L_k)$ 
5:   return LIETROTTER( $u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2}, \Delta t_k$ )  $\triangleright$  §2.6
6: end function

```

---

No constraint fixes the state at  $t_k + \Delta t_k/2$ , so the first scaling half step takes its ratios from the current power-law estimates, via PREDICTSCALE with  $i = 1/2$ :

$$\ln \frac{U_{k+1/2}}{U_k} = G_U(\frac{1}{2}, \beta_U; k), \quad \ln \frac{L_{k+1/2}}{L_k} = G_L(\frac{1}{2}, \beta_L; k). \quad (47)$$

The call to LIETROTTER then advances  $u_{k+1/2}$  over the full  $\Delta t_k$ , and its CONSTRAINEDSCALE substep — the second scaling half step — sets  $U_{k+1}/U_{k+1/2}$  and  $L_{k+1}/L_{k+1/2}$  so that  $u_{k+1}$  satisfies the constraints. The half-step state  $(u_{k+1/2}, \ln U_{k+1/2}, \ln L_{k+1/2})$  is internal to the step and is not stored. The top-level loop calls STRANG only when  $\beta_{\text{ok}}$  is true (§2.4), which guarantees that  $\beta_U, \beta_L$  were fitted to the stored history at the end of the previous step and that  $\ln \tau_{k-2}$  exists.

## 2.8 EstimateBeta: estimating $\beta_U$ and $\beta_L$

After every step with  $k+1 \geq 3$ , regardless of *scheme*,  $\beta_U$  and  $\beta_L$  are re-estimated from the log-scale history. Four stored states are needed:  $\ln X_k$  and  $\ln X_{k+1}$  for the ratio, and  $\ln \tau_{k-2:k}$  with the steps  $\Delta t_{k-2:k}$  for  $\Delta t_k^*$ . The update inverts the function  $G_X$  of (38) with  $i = 1$ , the same function that PREDICTSCALE evaluates forward with  $i = 1/2$ :

$$G_U(1, \beta_U; k) = \ln U_{k+1} - \ln U_k, \quad (48)$$

and identically with  $U \rightarrow L$  for  $\beta_L$ . The steps  $\Delta t_{k-2:k}$  end in the step just taken, so the  $\Delta t_k^*$  inverted here is the one PREDICTSCALE used for that step’s half step: under a varying step, prediction and estimation remain exact inverses on the same step.

The sensitivity of  $\beta_X$  to an error in  $\Delta t_k^*$  is weak. Taking logarithms of (38) for a small step,  $\ln(\ln X_{k+1} - \ln X_k) \approx \ln \beta_X + \ln \Delta t_k^* - \ln X_k / \beta_X$ , so a relative error  $\eta$  in  $\Delta t_k^*$  moves  $\beta_X$  by  $\delta \beta_X \approx \beta_X^2 \eta / \ln X_k$ , damped as the scales grow. It is not zero, however: feeding a stale step into (37) instead of  $\Delta t_k$  makes the estimate oscillate with the step sequence.

Each of (48) and its  $L$ -counterpart is a single scalar equation in a single unknown ( $\beta_U$ , resp.  $\beta_L$ ), solved by **brentq**; the implementer will need to choose a bracketing interval (e.g. centred on the previous  $\beta_U, \beta_L$  estimate — on  $\beta_U^{(0)}, \beta_L^{(0)}$  for the first update — widened adaptively if it fails to bracket a root).

If both root-finds succeed, ESTIMATEBETA returns the new  $\beta_U, \beta_L$  with  $\beta_{\text{ok}} = \text{true}$ . If either fails to bracket a root even after widening, the failed estimate keeps its previous value (which seeds the next bracket), and ESTIMATEBETA returns  $\beta_{\text{ok}} = \text{false}$ , so the next step falls back to LIETROTTER. Before  $k+1 \geq 3$  no update is made:  $\beta_U, \beta_L$  stay at  $\beta_U^{(0)}, \beta_L^{(0)}$  and  $\beta_{\text{ok}}$  stays false.

## 3 Post-processing

A completed run leaves behind the log-scale history  $\ln U_k, \ln L_k$  and the clock history  $\ln \tau_k$ . From these we recover the power-law parameters  $\alpha_X, \beta_X$  of the ansatz (33). This is the offline, global counterpart of ESTIMATEBETA (§2.8), which fits  $\beta_X$  one step at a time; here the whole history is used at once, so the fit also serves as a check on the run.

Write

$$\mathcal{L}_k = \ln L_k, \quad \mathcal{U}_k = \ln U_k, \quad (49)$$

together with the per-step log-ratios  $\ln \lambda_{X,k} = \ln X_k - \ln X_{k-1}$  of (18), which for a Strang step combine both half-step ratios. The stored scales are their cumulative sums,

$$\mathcal{L}_k = \mathcal{L}_0 + \sum_{i=1}^k \ln \lambda_{L,i}, \quad \mathcal{U}_k = \mathcal{U}_0 + \sum_{i=1}^k \ln \lambda_{U,i}, \quad (50)$$

which grow only linearly in  $k$ ; unlike  $L_k$  and  $U_k$  themselves,  $\mathcal{L}_k$  and  $\mathcal{U}_k$  never overflow.

### 3.1 The scaling rates under the ansatz

Differentiating the ansatz (33) with respect to  $t$ , and using the clock (3) to convert from  $t^*$ , the logarithmic rates (4) are

$$\sigma_X = \frac{1}{X} \frac{dX}{dt^*} \frac{dt^*}{dt} = \frac{\beta_X \tau}{t^* + \alpha_X}, \quad X \in \{U, L\}. \quad (51)$$

Inverting the ansatz to eliminate the unknown offset,

$$t^* + \alpha_X = X^{1/\beta_X}, \quad (52)$$

and substituting into (51) gives the rates in terms of the scales alone,

$$\sigma_X = \beta_X \tau X^{-1/\beta_X}. \quad (53)$$

Taking logarithms,

$$\ln \sigma_{X,k} = \ln \beta_X + \ln \tau_k - \frac{\mathcal{X}_k}{\beta_X}, \quad \mathcal{X}_k = \ln X_k. \quad (54)$$

Finally, the rates themselves are estimated directly from the log-ratios already in hand, without any finite difference of  $L_k$  or  $U_k$  themselves,

$$\sigma_{X,k} \approx \frac{\ln \lambda_{X,k+1}}{\Delta t_k}, \quad X \in \{U, L\}, \quad (55)$$

each divided by its own step, so that a run with a varying step (§2.3) is fitted exactly as a fixed-step one. The rates, not the log-ratios  $\ln \lambda_{X,k+1}$  themselves, are also what the run's diagnostics plot: a ratio scales with its step, so a CFL-driven dip in  $\Delta t_k$  would read as a dip in the scaling.

### 3.2 Recovering $\alpha_X$ and $\beta_X$

Equation (54) is linear in  $\mathcal{X}_k$  once the clock is divided out. Regressing

$$\ln \sigma_{X,k} - \ln \tau_k = \ln \beta_X - \frac{1}{\beta_X} \mathcal{X}_k \quad (56)$$

against  $\mathcal{X}_k$  therefore yields  $\beta_X$  twice over: once as  $e^{\text{intercept}}$  and once as  $-1/\text{slope}$ . Agreement between the two is a check that the run is in fact following a power law over the fitted range; disagreement indicates that it is not, or that the range includes the initial transient before the frame has locked on.

With  $\beta_X$  in hand, the offset follows from (52),

$$\alpha_X = X_k^{1/\beta_X} - t_k^*, \quad (57)$$

where the physical time is accumulated from the per-step increments of (36), with the weights (37) of the stored steps (the first two steps, which lack a three-point history, take  $\Delta t_i^* \approx \Delta t_i \tau_i$ ),

$$t_k^* = t_0^* + \sum_{i=0}^{k-1} \Delta t_i^*. \quad (58)$$

In exact arithmetic (57) is independent of  $k$ ; in practice it is averaged over the fitted range, and its scatter is a second check on the ansatz.

For the 1D diffusion equation the clock is  $\tau = L^2/D$ , and (54) specializes to

$$\ln \sigma_{L,k} = \ln \frac{\beta_L}{D} + \left(2 - \frac{1}{\beta_L}\right) \mathcal{L}_k, \quad (59)$$

$$\ln \sigma_{U,k} = \ln \frac{\beta_U}{D} + 2 \mathcal{L}_k - \frac{\mathcal{U}_k}{\beta_U}, \quad (60)$$

so that for  $X = L$  the regression (56) collapses onto a single variable, while for  $X = U$  it is a two-variable fit in  $\mathcal{L}_k$  and  $\mathcal{U}_k$ .

## 4 The 1D diffusion equation

The first instantiation of the framework is the one-dimensional diffusion equation,  $d = 1$ , with a single scalar component; we drop the component index throughout this section. It serves as a testbed: every operator the framework needs is available in closed form, so the only approximation in a run is the operator splitting itself.

### 4.1 Governing equation and clock

The dimensional equation is

$$\frac{\partial u^*}{\partial t^*} = D \frac{\partial^2 u^*}{\partial (x^*)^2}, \quad (61)$$

so that, in the notation of (1),

$$R^*(u^*) = D \frac{\partial^2 u^*}{\partial (x^*)^2}. \quad (62)$$

Under the rescaling (2) this term becomes

$$D \frac{\partial^2 (uU)}{\partial (x^*)^2} = DU \frac{\partial}{\partial x} \left( \frac{\partial u}{\partial x} \frac{1}{L} \right) \frac{1}{L} = \frac{DU}{L^2} \frac{\partial^2 u}{\partial x^2}, \quad (63)$$

having used  $dx/dx^* = 1/L$ . The clock is then chosen to make the physical term of the lemma (7) carry unit coefficient,

$$\frac{\tau}{U} \cdot \frac{DU}{L^2} = 1 \quad \implies \quad \tau = \frac{L^2}{D}, \quad (64)$$

which is the diffusive time scale of the current length scale, as expected. In log form, as required by (40),

$$\ln \tau_k = 2 \ln L_k - \ln D. \quad (65)$$

Substituting (64) into the rescaling lemma (7) gives the equation the algorithm integrates,

$$\frac{\partial u}{\partial t} = \sigma_L x \frac{\partial u}{\partial x} - \sigma_U u + \frac{\partial^2 u}{\partial x^2}. \quad (66)$$

### 4.2 The physics step

Because the diffusion term in (66) has unit coefficient, the physics operator (9),

$$\Phi_{\Delta t}^{(P)}(u_k) = u_k + \int_{t_k}^{t_k + \Delta t} \frac{\partial^2 u}{\partial x^2} dt, \quad (67)$$

has the closed-form spectral solution

$$\hat{u}_j^{(P)} = \hat{u}_j e^{-\kappa_j^2 \Delta t}, \quad \kappa_j = \frac{2\pi j}{H}, \quad j = 0, \dots, \lfloor N/2 \rfloor, \quad (68)$$

where  $\hat{u}_j = \text{RFFT}(u)_j$ . This is an exact one-shot spectral propagator, not an approximate time-marching scheme, so no sub-stepping or implicit solve is needed, and the accuracy requirement on EVOLVE (§2.5) is met trivially at any order. Inverse-transforming gives the advanced field  $u^{(P)}(x_n)$ .

### 4.3 Moment constraints

The constraint functionals of §1.4 are taken to be the zeroth and the normalized second moment of the field,

$$F_1[u] = \int u \, dx \equiv M_0, \quad F_2[u] = \frac{1}{M_0} \int x^2 u \, dx \equiv M_2, \quad (69)$$

where the  $M_0$  in  $F_2$  is the target value (43), a constant, not the zeroth moment of the field the functional is applied to. Thus  $M_0$  measures the amount of the conserved quantity and  $M_2$ , a mean-square width, its spread. Each functional is a single term of (29). Writing  $M_0^{\text{in}}, M_2^{\text{in}}$  for the moments of the field being scaled, and substituting  $y = x \lambda_L$  in each integral, the scaled field carries

$$\int \frac{1}{\lambda_U} u(x \lambda_L) \, dx = \frac{1}{\lambda_U \lambda_L} M_0^{\text{in}}, \quad (70)$$

$$\int x^2 \frac{1}{\lambda_U} u(x \lambda_L) \, dx = \frac{1}{\lambda_U \lambda_L^3} M_2^{\text{in}}, \quad (71)$$

so the homogeneity exponents of (29) are

$$(p_1, q_1) = (1, 1), \quad (p_2, q_2) = (1, 3), \quad (72)$$

with  $p_1 q_2 - p_2 q_1 = 2 \neq 0$ , so the constraint system (32) is uniquely solvable: the two moments do respond independently to amplitude and to length rescaling. Solving it,

$$\lambda_L = \sqrt{\frac{M_0 M_2^{\text{in}}}{M_0^{\text{in}} M_2}}, \quad (73)$$

$$\lambda_U = \frac{M_0^{\text{in}}}{M_0 \lambda_L} = \left( \frac{M_0^{\text{in}}}{M_0} \right)^{3/2} \sqrt{\frac{M_2}{M_2^{\text{in}}}}. \quad (74)$$

Because the normalization of  $F_2$  is a constant, these ratios are the same as for the raw second moment  $\int x^2 u \, dx$ ; written in terms of the raw moments,  $M_0$  cancels from  $M_2^{\text{in}}/M_2$ .

### 4.4 Spectral interpolation on the real transform

For  $d = 1$  and a real-valued field, the generic interpolation formula (27) is realized on the non-negative frequency components of a real Fast Fourier Transform (RFFT):

$$\begin{aligned} \Phi_{\Delta t}^{(S)}(u_k)(x) = \frac{1}{\lambda_{U,k+1}} & \left\{ \hat{u}_0^{(k)} + 2 \sum_{j=1}^{N/2-1} (-1)^j \left[ \text{Re}(\hat{u}_j^{(k)}) \cos\left(\frac{2\pi j \lambda_{L,k+1} x}{H}\right) \right. \right. \\ & \left. \left. - \text{Im}(\hat{u}_j^{(k)}) \sin\left(\frac{2\pi j \lambda_{L,k+1} x}{H}\right) \right] \right. \\ & \left. + (-1)^{N/2} \hat{u}_{N/2}^{(k)} \cos\left(\frac{\pi N \lambda_{L,k+1} x}{H}\right) \right\}, \end{aligned} \quad (75)$$

where  $\hat{u}_j^{(k)} = \frac{1}{N} \text{RFFT}(u_k)_j$  for  $j = 0, 1, \dots, N/2$ , and the Nyquist term at  $j = N/2$  is included only if  $N$  is even. The factor of 2 accounts for the Hermitian conjugate modes omitted by the real transform. (75) holds for  $|\lambda_{L,k+1} x| \leq H/2$ ; outside that the cutoff (28) sets  $\Phi_{\Delta t}^{(S)}(u_k)(x) = 0$ .

The alternating sign  $(-1)^j$  is the phase-origin correction, and it is not optional. The transform phases its coefficients to the sample index  $n$ , whose origin  $n = 0$  sits at  $x = -H/2$  on the centred grid (41) rather than at  $x = 0$ . Since the scaling operator (19) dilates about the centre of the domain, each mode must be evaluated at  $\lambda_{L,k+1} x + H/2$  — dilate about  $x = 0$ , then shift to the transform's phase origin — and not at  $\lambda_{L,k+1}(x + H/2)$ , which dilates the shift along with the coordinate and agrees with it only at  $\lambda_{L,k+1} = 1$ . As  $H/2$  is half a period, that shift contributes the constant phase  $\exp(i\kappa_j H/2) = (-1)^j$ . Dropping it altogether returns  $u_k(\lambda_{L,k+1} x - H/2)$ , the wanted profile displaced by half the domain, rather than the  $u_k(\lambda_{L,k+1} x)$  that (19) calls for. The same factor carries to each axis in three dimensions (§5.10).

## 4.5 Boundary treatment

The substep APPLYBC of §2.5 imposes a homogeneous Dirichlet condition by clamping the two edge grid points of the centred domain (41) after each completed step,

$$u_{k+1}(x_0) \leftarrow 0, \quad u_{k+1}(x_{N-1}) \leftarrow 0. \quad (76)$$

On the centred grid  $x \in [-H/2, H/2]$  these are two distinct grid points, not periodic images of one another.

*Open.* This sits in tension with the rest of the discretization: the spectral propagator (68) presumes a periodic field, whereas (76) imposes a Dirichlet condition. The spectral interpolation (75) no longer does at the edges: with the cutoff (28) it treats everything beyond the box as zero, which is consistent with the clamp, but it is still a periodic interpolant inside the box. The two are consistent only so long as  $u$  is negligible near the domain edges, which requires the initial condition to be localized well inside the domain and the run to be stopped before the profile spreads to the boundary. The scaling step works against this — it is what keeps the profile from spreading in the self-similar frame — but the condition is an assumption of the scheme rather than a consequence of it, and is not currently monitored.

## 5 The Navier–Stokes equations

The second instantiation is the incompressible Navier–Stokes equations, with  $d = 3$  and  $u_i$  the velocity field.

### 5.1 Governing equations and clock

The equations are solved as a large-eddy simulation (LES). Let  $\mathcal{G}_{\Delta^*}$  denote a spatial filter of width  $\Delta^*$ . A tilde marks a quantity of the filtered (resolved) field, and the same symbol without a tilde the corresponding quantity of the full field:  $\tilde{u}_i^* = \mathcal{G}_{\Delta^*}(u_i^*)$  and  $\tilde{p}^*$  are the filtered velocity and pressure, while  $u_i^*$  and  $p^*$  are the full ones. For a quantity derived from the velocity, the tilde means it is evaluated from  $\tilde{u}_i^*$ , not that it is filtered; thus  $\frac{1}{2}\tilde{u}_i^*\tilde{u}_i^*$  is not  $\mathcal{G}_{\Delta^*}(\frac{1}{2}u_i^*u_i^*)$ . The effect of the unresolved scales enters through the kinematic subgrid-scale stress

$$\tau_{ij}^* = \mathcal{G}_{\Delta^*}(u_i^*u_j^*) - \tilde{u}_i^*\tilde{u}_j^*, \quad (77)$$

which, like the eddy viscosity below, exists only in the LES and carries no tilde. The dimensional filtered equations are

$$\frac{\partial \tilde{u}_i^*}{\partial t^*} + \tilde{u}_j^* \frac{\partial \tilde{u}_i^*}{\partial x_j^*} = -\frac{1}{\rho} \frac{\partial \tilde{p}^*}{\partial x_i^*} + \nu \frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}, \quad (78)$$

together with the continuity equation,

$$\frac{\partial \tilde{u}_i^*}{\partial x_i^*} = 0, \quad (79)$$

so that, in the notation of (1),

$$R_i^*(\tilde{u}^*) = -\tilde{u}_j^* \frac{\partial \tilde{u}_i^*}{\partial x_j^*} - \frac{1}{\rho} \frac{\partial \tilde{p}^*}{\partial x_i^*} + \nu \frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} - \frac{\partial \tau_{ij}^*}{\partial x_j^*}. \quad (80)$$

In addition to the rescaling (2) of  $\tilde{u}_i$  and  $x_j$ , the pressure is nondimensionalized by the dynamic pressure,

$$\tilde{p} = \frac{\tilde{p}^*}{\rho U^2}, \quad (81)$$

and the subgrid-scale stress, which carries units of velocity squared, by  $U^2$ ,

$$\tau_{ij} = \frac{\tau_{ij}^*}{U^2}. \quad (82)$$

With the filter width nondimensionalized as  $\Delta = \Delta^*/L$ , the definitions carry over unchanged,  $\tilde{u}_i = \mathcal{G}_{\Delta}(u_i)$  and  $\tau_{ij} = \mathcal{G}_{\Delta}(u_i u_j) - \tilde{u}_i \tilde{u}_j$ . Term by term, the convective, pressure, and subgrid-scale terms scale as  $U^2/L$ ,

$$\tilde{u}_j^* \frac{\partial \tilde{u}_i^*}{\partial x_j^*} = \frac{U^2}{L} \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j}, \quad \frac{1}{\rho} \frac{\partial \tilde{p}^*}{\partial x_i^*} = \frac{U^2}{L} \frac{\partial \tilde{p}}{\partial x_i}, \quad \frac{\partial \tau_{ij}^*}{\partial x_j^*} = \frac{U^2}{L} \frac{\partial \tau_{ij}}{\partial x_j}, \quad (83)$$

while the viscous term scales as  $\nu U/L^2$ ,

$$\nu \frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*} = \frac{\nu U}{L^2} \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j}. \quad (84)$$

The clock is chosen as the inverse of the advection timescale

$$\tau = \frac{L}{U}. \quad (85)$$

In log form, as required by (40),

$$\ln \tau_k = \ln L_k - \ln U_k. \quad (86)$$

Applying  $\tau/U = L/U^2$  to (80) and introducing the instantaneous Reynolds number

$$Re = \frac{UL}{\nu}, \quad (87)$$

the rescaling lemma (7) gives the nondimensional momentum equation

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L x_j \frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U \tilde{u}_i - \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{1}{Re} \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j}, \quad (88)$$

and the continuity equation is unchanged by the rescaling,

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0. \quad (89)$$

## 5.2 Subgrid stress model

The subgrid-scale stress  $\tau_{ij}^*$  of (78) is closed with an eddy-viscosity model. Its deviatoric part is taken proportional to the resolved strain rate,

$$\tau_{ij}^* - \frac{1}{3} \tau_{kk}^* \delta_{ij} = -2\nu_t^* \tilde{S}_{ij}^*, \quad \tilde{S}_{ij}^* = \frac{1}{2} \left( \frac{\partial \tilde{u}_i^*}{\partial x_j^*} + \frac{\partial \tilde{u}_j^*}{\partial x_i^*} \right), \quad (90)$$

with the isotropic part  $\frac{1}{3} \tau_{kk}^* \delta_{ij}$  absorbed into the pressure. The eddy viscosity  $\nu_t^*$  carries the units of  $UL$ , so it is nondimensionalized as  $\nu_t = \nu_t^*/(UL)$ . Noting that  $\tilde{S}_{ij}^* = (U/L) \tilde{S}_{ij}$ , the nondimensional subgrid stress (82) is

$$\tau_{ij} - \frac{1}{3} \tau_{kk} \delta_{ij} = -2\nu_t \tilde{S}_{ij}. \quad (91)$$

The models below differ only in the choice of  $\nu_t^*$ .

### 5.2.1 Smagorinsky model

The standard Smagorinsky model takes the eddy viscosity

$$\nu_t^* = (C_s \Delta^*)^2 |\tilde{S}^*|, \quad |\tilde{S}^*| = \sqrt{2 \tilde{S}_{ij}^* \tilde{S}_{ij}^*}, \quad (92)$$

where  $C_s$  is the Smagorinsky constant and  $\Delta^*$  is the filter width. With the nondimensional filter width  $\Delta = \Delta^*/L$  taken equal to the grid spacing,  $\Delta = \Delta x$ , the eddy viscosity becomes

$$\nu_t = UL (C_s \Delta)^2 |\tilde{S}|, \quad \nu_t = (C_s \Delta)^2 |\tilde{S}|. \quad (93)$$

### 5.2.2 Self-similar constant eddy viscosity

The second model keeps the dependence on the filter width of (92) but replaces the local strain rate  $|\tilde{S}^*|$  by the strain-rate scale  $U/L$  of the self-similar frame,

$$\nu_t^* = C_\nu (\Delta^*)^2 \frac{U}{L}, \quad \nu_t = C_\nu \Delta^2, \quad (94)$$

with  $C_\nu$  a constant and  $\Delta = \Delta^*/L = \Delta x$  as in the Smagorinsky model. The eddy viscosity is uniform in space, so with (79) the subgrid stress term reduces to a Laplacian,

$$-\frac{\partial \tau_{ij}^*}{\partial x_j^*} = \nu_t^* \frac{\partial^2 \tilde{u}_i^*}{\partial x_j^* \partial x_j^*}, \quad -\frac{\partial \tau_{ij}}{\partial x_j} = C_\nu \Delta^2 \frac{\partial^2 \tilde{u}_i}{\partial x_j \partial x_j}. \quad (95)$$

In the self-similar frame the model is therefore a constant viscosity, of the same form as the molecular term of (88) with an effective Reynolds number  $Re_t = 1/(C_\nu \Delta^2)$ . Unlike  $Re$ , which changes with  $UL$  as the flow evolves,  $Re_t$  is fixed for the run, while in physical variables, where  $\Delta^* = L\Delta$  grows with the flow,  $\nu_t^*$  follows the product  $UL$ .

### The momentum equation

In the limit of infinite Reynolds number,  $1/Re = 0$ , and with either model the momentum equation (88) becomes

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L x_j \frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U \tilde{u}_i - \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[ 2\nu_t \tilde{S}_{ij} \right], \quad (96)$$

where  $\tilde{p}$  now denotes the modified pressure that includes  $\frac{1}{3}\tau_{kk}$ , and  $\nu_t$  is given by (93) or (94). For the self-similar constant eddy viscosity the last term is  $C_\nu \Delta^2 \partial^2 \tilde{u}_i / \partial x_j \partial x_j$ , by (95).

### 5.3 Sponge forcing

The true boundary conditions we would like to enforce is  $\tilde{u}_i = 0$  at the edge of the domain. However, we are using a pseudo-spectral solver so this boundary condition cannot be directly enforced. To solve this we introduce a fringe region at the edge of the domain [1], which introduces a forcing term to the right hand side of the Navier–Stokes equation

$$f_i = -W(r) \tilde{u}_i, \quad (97)$$

where  $r = \sqrt{x_i x_i}$  is the distance from the origin and  $W(r)$  is a weighting function that increases with increasing  $r$ .

The weighting function is

$$W(r) = W_{\max} S\left(\frac{r - r_0}{r_1 - r_0}\right), \quad (98)$$

where the smooth step  $S$  is

$$S(\xi) = \begin{cases} 0, & \xi \leq 0, \\ \frac{1}{1 + \exp\left(\frac{1}{\xi - 1} + \frac{1}{\xi}\right)}, & 0 < \xi < 1, \\ 1, & \xi \geq 1. \end{cases} \quad (99)$$

The function  $S$  is  $C^\infty$  on the whole real line. Every derivative of it vanishes at  $\xi = 0$  and at  $\xi = 1$ , it rises monotonically between them, and it is antisymmetric about its midpoint,  $S(1 - \xi) = 1 - S(\xi)$ , so that  $S(1/2) = 1/2$ . The weight (98) is therefore exactly zero in the sphere  $r \leq r_0$ , rises smoothly across the shell  $r_0 < r < r_1$ , and is exactly  $W_{\max}$  for  $r \geq r_1$ . Adding (97) to (96), the momentum equation actually solved is

$$\frac{\partial \tilde{u}_i}{\partial t} = \sigma_L x_j \frac{\partial \tilde{u}_i}{\partial x_j} - \sigma_U \tilde{u}_i - \tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j} - \frac{\partial \tilde{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[ 2\nu_t \tilde{S}_{ij} \right] - W(r) \tilde{u}_i. \quad (100)$$

## 5.4 Fourier representation

Using the Fourier convention (20), the pressure gradient becomes

$$\widehat{\frac{\partial \bar{p}}{\partial x_i}} = i\kappa_i \hat{p}. \quad (101)$$

The convective term

$$\hat{N}_i = \widehat{\tilde{u}_j \frac{\partial \tilde{u}_i}{\partial x_j}}, \quad (102)$$

will be evaluated pseudo-spectrally. How the subgrid stress term

$$\hat{G}_i = \frac{\partial}{\partial x_j} \widehat{[2\nu_t \tilde{S}_{ij}]} \quad (103)$$

is evaluated depends on the model of §5.2. In both cases the strain rate is available spectrally,

$$\hat{\tilde{S}}_{ij} = \frac{i}{2} \left( \kappa_j \hat{u}_i + \kappa_i \hat{u}_j \right). \quad (104)$$

**Smagorinsky model.** The eddy viscosity (93) depends on the field, so  $\hat{G}_i$  is evaluated pseudo-spectrally:  $\tilde{S}_{ij}$  is transformed to physical space, the products  $|\tilde{S}|$ ,  $\nu_t = (C_s \Delta)^2 |\tilde{S}|$  and  $T_{ij} = 2\nu_t \tilde{S}_{ij}$  are formed there, and the divergence is taken spectrally,

$$\hat{G}_i = i\kappa_j \hat{T}_{ij}. \quad (105)$$

Since  $|\tilde{S}|$  is not a polynomial in  $\tilde{u}$ ,  $\hat{G}_i$  carries aliasing that the 2/3 rule does not remove entirely.

**Self-similar constant eddy viscosity.** The eddy viscosity (94) is uniform, and by (95)

$$\hat{G}_i = -C_\nu \Delta^2 \kappa^2 \hat{u}_i. \quad (106)$$

This is exact and diagonal in  $\kappa$ : no product is formed in physical space, so  $\hat{G}_i$  needs no transforms, carries no aliasing, and stays in the retained band. By (109) it is also solenoidal,  $\kappa_i \hat{G}_i = 0$ , so it contributes nothing to the pressure below and passes through the projection (115) unchanged.

**Sponge forcing.** The sponge forcing (97) is formed as a product in physical space and transformed,

$$\hat{F}_i = \widehat{W \tilde{u}_i}, \quad (107)$$

with  $W$  tabulated once on the grid (41). Because  $W$  is smooth but not band-limited,  $\hat{F}_i$  also carries content beyond the retained band, which the 2/3 mask removes along with that of  $\hat{N}_i$  and, under the Smagorinsky model,  $\hat{G}_i$ . The scaling terms transform as in (23) with  $d = 3$ , so the transform of (100) is

$$\frac{\partial \hat{u}_i}{\partial t} = -\sigma_L \left( \kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + 3 \hat{u}_i \right) - \sigma_U \hat{u}_i - \hat{N}_i - i\kappa_i \hat{p} + \hat{G}_i - \hat{F}_i, \quad (108)$$

and the transform of the continuity equation (89) is

$$\kappa_i \hat{u}_i = 0. \quad (109)$$

## 5.5 Pressure and the solenoidal projection

Contracting (108) with  $\kappa_i$  and using the continuity constraint gives

$$0 = -\sigma_L \left( \kappa_i \kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + d \kappa_i \hat{u}_i \right) - \sigma_U \kappa_i \hat{u}_i - \kappa_i \hat{N}_i - i\kappa^2 \hat{p} + \kappa_i \hat{G}_i - \kappa_i \hat{F}_i. \quad (110)$$

The remaining scaling contribution also vanishes: differentiating (109) with respect to  $\kappa_j$  gives

$$\frac{\partial}{\partial \kappa_j} \left( \kappa_i \hat{u}_i \right) = \hat{u}_j + \kappa_i \frac{\partial \hat{u}_i}{\partial \kappa_j} = 0 \quad \implies \quad \kappa_i \frac{\partial \hat{u}_i}{\partial \kappa_j} = -\hat{u}_j, \quad (111)$$

so that

$$\kappa_i \kappa_j \partial \hat{u}_i / \partial \kappa_j = -\kappa_j \hat{u}_j = 0. \quad (112)$$

Therefore, the equation for pressure is

$$\mathbf{i} \kappa^2 \hat{p} = -\kappa_i \left( \hat{N}_i - \hat{G}_i + \hat{F}_i \right) \quad \implies \quad \hat{p} = \mathbf{i} \frac{\kappa_l \left( \hat{N}_l - \hat{G}_l + \hat{F}_l \right)}{\kappa^2}, \quad \kappa \neq 0, \quad (113)$$

with  $\hat{p}$  at  $\kappa = 0$  fixed arbitrarily, since only its gradient enters the momentum equation. Substituting (113) into (108), the pressure term becomes

$$-\mathbf{i} \kappa_i \hat{p} = \frac{\kappa_i \kappa_l}{\kappa^2} \left( \hat{N}_l - \hat{G}_l + \hat{F}_l \right), \quad (114)$$

which combines with  $-\hat{N}_i + \hat{G}_i - \hat{F}_i$  to give the solenoidal projection of the convective, subgrid stress, and sponge terms together. Defining the projection tensor

$$P_{il}(\boldsymbol{\kappa}) = \delta_{il} - \frac{\kappa_i \kappa_l}{\kappa^2}, \quad (115)$$

the evolution equation in fourier space is (7),

$$\frac{\partial \hat{u}_i}{\partial t} = \underbrace{-\sigma_L \left( \kappa_j \frac{\partial \hat{u}_i}{\partial \kappa_j} + 3 \hat{u}_i \right)}_{\text{scaling terms}} - \underbrace{\sigma_U \hat{u}_i - P_{il}(\boldsymbol{\kappa}) \left( \hat{N}_l - \hat{G}_l + \hat{F}_l \right)}_{\text{physical terms}}. \quad (116)$$

## 5.6 The physics step

The physics operator (9) is the flow of the bracketed physical terms of (116),

$$\Phi_{\Delta t}^{(P)}(\hat{u}_i(t)) = \hat{u}_i(t) - \int_t^{t+\Delta t} P_{il}(\boldsymbol{\kappa}) \left( \hat{N}_l - \hat{G}_l + \hat{F}_l \right) dt. \quad (117)$$

We take  $\Phi^{(P)}$  to be a single step of the classical fourth-order Runge–Kutta method (RK4) applied to the full physical right-hand side, the subgrid stress term  $\hat{G}_i$  and the sponge forcing  $\hat{F}_i$  marched together with the convection. It is set out in Appendix A, together with the stability limits it places on  $\Delta t$ . The advective limit depends on the state, and is enforced by the CFL time step of §2.3, through MAXSTABLEDT (170). The sponge limit and, under the self-similar constant eddy viscosity, the diffusive limit are fixed by the run parameters and are assumed rather than enforced.

## 5.7 Post-processing

We extract two point statistics from the runs, the dissipation rate and the kinetic energy (§5.7.2), both at the origin. Each is a dimensional prefactor built from  $U$  and  $L$  times a nondimensional quantity computed from the self-similar field, so both follow from the same scaling argument, which §5.7.1 sets out once. The split of the kinetic energy into resolved and subgrid parts set out in §5.7.2 also supplies the integrand of the constraint functionals of §5.8, which is why this subsection precedes them.

### 5.7.1 Scaling of point statistics at the origin

Let  $Q^*$  be a pointwise quantity that nondimensionalizes as

$$Q^* = U^{m_U} L^{m_L} Q, \quad (118)$$

where the nondimensional  $Q$  is evaluated from the self-similar fields. We are interested in  $Q^*$  at the origin,  $\mathbf{x}^* = \mathbf{x} = 0$ , which is a fixed point of the rescaling (2) and a grid point of (41) ( $n = N/2$  for even  $N$ ), so no interpolation is required. Substituting the power-law ansatz (33) for  $U$  and  $L$ ,

$$Q^*(0, t^*) = (t^* + \alpha_U)^{m_U \beta_U} (t^* + \alpha_L)^{m_L \beta_L} Q(0, t). \quad (119)$$

**Scaling of the ensemble average.** Let  $\overline{\cdot}$  denote the average over an ensemble of initial conditions. We make two assumptions:

1. The parameters  $\alpha_U, \beta_U, \alpha_L, \beta_L$  are the same for every member of the ensemble, i.e.  $U(t^*)$  and  $L(t^*)$  are invariant to the initial condition.
2. In the self-similar frame the flow is statistically stationary, so that  $\overline{Q(0, t)}$  is independent of  $t$ .

Under these assumptions

$$\overline{Q^*}(0, t^*) = (t^* + \alpha_U)^{m_U \beta_U} (t^* + \alpha_L)^{m_L \beta_L} \overline{Q(0)}. \quad (120)$$

If in addition we assume that the system is ergodic the ensemble average is equivalent to time averaging and we have

$$\overline{Q(0)} = \frac{1}{T} \int_0^T Q(0, t) dt, \quad (121)$$

and it is enough to look at a single long time horizon simulation. In practice  $T$  is finite and the initial transient must be discarded, so (121) is only an estimate of  $\overline{Q(0)}$ ; Appendix B describes how its statistical error is computed.

### 5.7.2 Dissipation rate and kinetic energy

The dissipation rate is not resolved by the LES, and in the limit  $1/Re = 0$  its resolved part vanishes. Assuming the subgrid scales are in local equilibrium, we take it equal to the transfer of energy from resolved to subgrid scales [2],

$$\Pi^* = -\tau_{ij}^* \tilde{S}_{ij}^* = \nu_t^* |\tilde{S}^*|^2, \quad \overline{\epsilon^*} = \overline{\Pi^*}, \quad (122)$$

$$\Pi^* = \frac{U^3}{L} \Pi, \quad \Pi = \nu_t |\tilde{S}|^2 = \begin{cases} (C_s \Delta)^2 |\tilde{S}|^3, & \text{Smagorinsky model,} \\ C_\nu \Delta^2 |\tilde{S}|^2, & \text{self-similar constant eddy viscosity,} \end{cases} \quad (123)$$

by (93) and (94).

The resolved kinetic energy is

$$\tilde{k}^* = U^2 \tilde{k}, \quad \tilde{k} = \frac{1}{2} \tilde{u}_i \tilde{u}_i, \quad (124)$$

and contracting (77) gives the filtered kinetic energy of the full field [2],

$$\mathcal{G}_\Delta(k) = \tilde{k} + k_{\text{sgs}}, \quad k_{\text{sgs}} = \frac{1}{2} \tau_{kk}. \quad (125)$$

The isotropic part of  $\tau_{ij}$  is absorbed into the pressure, so  $k_{\text{sgs}}$  is not computed and must be estimated. For both subgrid stress models we estimate it with the Smagorinsky eddy viscosity (93), so that the estimate is the same function of the resolved field in either case. Equating the production  $\nu_t |\tilde{S}|^2 = C_s^2 \Delta^2 |\tilde{S}|^3$  to the dissipation  $C_\epsilon k_{\text{sgs}}^{3/2} / \Delta$ , with  $C_\epsilon$  a constant of order one,

$$k_{\text{sgs}} = \left( \frac{C_s^2}{C_\epsilon} \right)^{2/3} \Delta^2 |\tilde{S}|^2, \quad (126)$$

with  $k_{\text{sgs}}^* = U^2 k_{\text{sgs}}$ .

## 5.8 Constraint functionals

The length and velocity scales will be set such that the zeroth and normalized second radial moments of the energy density  $e = \tilde{u}_i \tilde{u}_i + 2k_{\text{sgs}} = 2\mathcal{G}_\Delta(k)$ , twice the filtered kinetic energy per unit mass of the full field (125), are constant. Mathematically, these moments are defined as

$$F_1[\tilde{u}] = \int (\tilde{u}_i \tilde{u}_i + 2k_{\text{sgs}}) d\mathbf{x} \equiv E_0, \quad F_2[\tilde{u}] = \frac{1}{E_0} \int r^2 (\tilde{u}_i \tilde{u}_i + 2k_{\text{sgs}}) d\mathbf{x} \equiv E_2, \quad (127)$$

where  $r^2 = x_j x_j$  is the squared distance from the centre of the domain (41), and the  $E_0$  in  $F_2$  is the target value (43), a constant. A normalized filter preserves volume integrals, so  $\int \mathcal{G}_\Delta(k) d\mathbf{x} = \int k d\mathbf{x}$  and  $E_0$  is twice the total kinetic energy of the full field, resolved and unresolved, not only of the resolved field.  $E_2$  is the energy-weighted mean square radius of  $\mathcal{G}_\Delta(k)$ ; for an isotropic filter of variance  $\sigma_\Delta^2$  per direction it exceeds that of  $k$  itself by the constant  $3\sigma_\Delta^2$ , of order  $\Delta^2$ . As in the 1D case,  $E_0$  measures the amplitude of the field and  $E_2$  its spread.

The subgrid energy is evaluated pointwise from the estimate (126), which is the same for both models,

$$k_{\text{sgs}} = c_k \Delta^2 |\tilde{S}|^2, \quad c_k = \left( \frac{C_s^2}{C_\epsilon} \right)^{2/3}, \quad (128)$$

so  $C_s$  and  $C_\epsilon$  are parameters of the run, not only of the post-processing, also under the self-similar constant eddy viscosity.  $|\tilde{S}|$  is formed in physical space from the strain (104) of the field being constrained, at the cost of six inverse transforms per CONSTRAINEDSCALE; EVOLVE evaluates the strain only at its own stages.

Each functional splits into a resolved and a subgrid term of (29),

$$F_{11} = \int \tilde{u}_i \tilde{u}_i d\mathbf{x}, \quad F_{12} = \int 2k_{\text{sgs}} d\mathbf{x}, \quad F_{21} = \frac{1}{E_0} \int r^2 \tilde{u}_i \tilde{u}_i d\mathbf{x}, \quad F_{22} = \frac{1}{E_0} \int r^2 2k_{\text{sgs}} d\mathbf{x}. \quad (129)$$

Writing  $F_{mj}^{\text{in}}$  for the values carried by the field being scaled and substituting  $\mathbf{y} = \mathbf{x} \lambda_L$  in each integral, with  $d = 3$ , the resolved terms give

$$\int \frac{1}{\lambda_U^2} \tilde{u}_i \tilde{u}_i(\mathbf{x} \lambda_L) d\mathbf{x} = \frac{1}{\lambda_U^2 \lambda_L^3} F_{11}^{\text{in}}, \quad (130)$$

$$\frac{1}{E_0} \int r^2 \frac{1}{\lambda_U^2} \tilde{u}_i \tilde{u}_i(\mathbf{x} \lambda_L) d\mathbf{x} = \frac{1}{\lambda_U^2 \lambda_L^5} F_{21}^{\text{in}}. \quad (131)$$

The filter width  $\Delta$  is fixed in the self-similar frame, so it does not scale with the field. The strain of the scaled field is  $(\lambda_L/\lambda_U) \tilde{S}_{ij}(\mathbf{x} \lambda_L)$ , and by (128) its subgrid energy is  $(\lambda_L/\lambda_U)^2 k_{\text{sgs}}(\mathbf{x} \lambda_L)$ , so the subgrid terms give

$$F_{12} \rightarrow \lambda_U^{-2} \lambda_L^{-1} F_{12}^{\text{in}}, \quad F_{22} \rightarrow \lambda_U^{-2} \lambda_L^{-3} F_{22}^{\text{in}}. \quad (132)$$

The homogeneity exponents of (29) are therefore

$$(p_{11}, q_{11}) = (2, 3), \quad (p_{12}, q_{12}) = (2, 1), \quad (p_{21}, q_{21}) = (2, 5), \quad (p_{22}, q_{22}) = (2, 3). \quad (133)$$

The resolved and subgrid terms scale differently, so the constraints are no longer the linear system (32) and are solved through the residuals (30).

**Solvability.** Write  $w_1 = w_{11}$  and  $w_2 = w_{21}$  for the resolved fractions of the two functionals in (31);  $w_1$  is the volume-integrated form of the resolution criterion of §5.7.2. The weighted exponents are  $\bar{p}_m = 2$ ,  $\bar{q}_1 = 1 + 2w_1$  and  $\bar{q}_2 = 3 + 2w_2$ , so the determinant is  $\bar{p}_1 \bar{q}_2 - \bar{p}_2 \bar{q}_1 = 4 + 4(w_2 - w_1)$ , which on  $[0, 1]^2$  vanishes only at  $(w_1, w_2) = (1, 0)$ . That corner is not attained:  $k_{\text{sgs}} \geq 0$ , so  $w_1 = 1$  forces  $k_{\text{sgs}} = 0$  and with it  $w_2 = 1$ . The Jacobian is therefore nonsingular at every iterate, and the Newton iteration of §1.4 is well defined.

**Closed form.** All four terms share  $p = 2$ , so  $\lambda_U$  factors out of both functionals. Their ratio fixes  $\lambda_L$  alone, and with  $\mu = \lambda_L^2$  it is the quadratic

$$E_2 F_{12}^{\text{in}} \mu^2 + B \mu - E_0 F_{21}^{\text{in}} = 0, \quad B = E_2 F_{11}^{\text{in}} - E_0 F_{22}^{\text{in}}. \quad (134)$$

Its coefficients have opposite signs at the two ends, so it has exactly one positive root,

$$\lambda_L^2 = \frac{2E_0 F_{21}^{\text{in}}}{B + \sqrt{B^2 + 4E_0 E_2 F_{12}^{\text{in}} F_{21}^{\text{in}}}}, \quad (135)$$

written in the form that is free of cancellation for  $B \geq 0$ , which holds whenever more than half of each moment is resolved. The first functional then gives

$$\lambda_U^2 = \frac{F_{11}^{\text{in}} \lambda_L^{-3} + F_{12}^{\text{in}} \lambda_L^{-1}}{E_0}. \quad (136)$$

For  $F_{12}^{\text{in}} = F_{22}^{\text{in}} = 0$  these reduce to  $\lambda_L^2 = E_0 F_{21}^{\text{in}} / (E_2 F_{11}^{\text{in}})$  and  $\lambda_U^2 = F_{11}^{\text{in}} \lambda_L^{-3} / E_0$ , the solution of the linear system (32) for the resolved energy alone, whose length ratio is identical in form to its 1D counterpart (73).

## 5.9 Initial condition

Because the curl of a vector field is divergence-free we will define  $\tilde{u}_i$  as the curl of some field  $A_i$

$$\tilde{u}_i = \varepsilon_{ijk} \frac{\partial A_k}{\partial x_j}, \quad \text{equivalently} \quad \hat{u}_i = i \varepsilon_{ijk} \kappa_j \hat{A}_k. \quad (137)$$

$\tilde{u}_i$  is built on the grid (41) in four steps, followed by a rescaling.

1. *Noise.* Each component  $A_k(\mathbf{x}_n)$  is an independent standard normal, drawn from a seeded Mersenne Twister.
2. *Spectral filter.* The noise is given a spectrum,

$$\hat{A}_k \leftarrow \hat{A}_k e^{-\kappa^2 / \kappa_p^2}. \quad (138)$$

3. *Window.* The potential is localized in physical space,

$$A_k \leftarrow A_k e^{-r^2 / 2\sigma^2}, \quad (139)$$

with  $r$  the same distance from the centre of the domain that  $E_2$  uses in (127).

4. *Rescaling.* Finally  $\tilde{u}_i \leftarrow A \tilde{u}_i$ , with the amplitude  $A > 0$  chosen so that the field delivered to the integrator carries a prescribed  $E_0$ . By (128) both terms of  $F_1$  scale as  $A^2$ ,  $F_1[A\tilde{u}] = A^2 (F_{11}[\tilde{u}] + F_{12}[\tilde{u}])$ , with both terms evaluated as the Riemann sum (42), so  $A = \sqrt{E_0 / F_1[\tilde{u}]}$ .

## 5.10 Spectral interpolation in three dimensions

The scaling operator (19) needs the same band-limited evaluation (27) as in one dimension, but at  $d = 3$  a direct evaluation of it is no longer affordable, and its realization on the real transform differs enough from (75) to be worth writing out.

**The sum on the real transform.** Write  $\hat{u}_i^{(k)}(\mathbf{j}) = N^{-3} \text{RFFT}(\tilde{u}_k)_j$  for the normalized coefficients, with the first axis carrying only the non-negative frequencies  $j_1 = 0, 1, \dots, N/2$  and the remaining two carrying the full signed set  $j_d = 0, 1, \dots, N/2, -(N/2 - 1), \dots, -1$ . Two corrections turn (27) into a sum over that layout. First, the omitted Hermitian conjugates are restored by the multiplicity

$$m_{j_1} = \begin{cases} 1, & j_1 = 0, \text{ or } j_1 = N/2 \text{ with } N \text{ even,} \\ 2, & \text{otherwise,} \end{cases} \quad (140)$$

exactly as the factor of 2 does in (75). Second, the transform phases its coefficients to the sample index  $n$ , which on the centred grid (41) places its origin at  $x = -H/2$  rather than at  $x = 0$ . Since the scaling operator dilates about the centre of the domain, each mode must be evaluated at  $\lambda_L x + H/2$ , not at  $\lambda_L(x + H/2)$  — the two agree only at  $\lambda_L = 1$ . As  $H/2$  is half a period, the correction is the constant phase  $\exp(i\kappa_j H/2) = (-1)^j$  per axis. Together,

$$\Phi_{\Delta t}^{(S)}(\tilde{u}_k)_i(\mathbf{x}) = \frac{1}{\lambda_{L,k+1}} \operatorname{Re} \sum_{j_1=0}^{N/2} \sum_{j_2} \sum_{j_3} m_{j_1} \hat{\tilde{u}}_i^{(k)}(\mathbf{j}) \prod_{d=1}^3 (-1)^{j_d} \exp\left(\frac{2\pi i j_d \lambda_{L,k+1} x_d}{H}\right). \quad (141)$$

**Separable evaluation.** Evaluated as written, (141) costs  $O(N^{2d})$ : every retained mode is summed at every grid point, which at  $N = 128$  is  $4 \times 10^{12}$  complex multiply-adds per scaling substep. The kernel, however, is a product of one factor per axis, so the triple sum separates and can be applied one axis at a time. Define the two interpolation matrices

$$W_{n j_1}^{(1)} = m_{j_1} (-1)^{j_1} \exp\left(\frac{2\pi i j_1 \lambda_{L,k+1} x_n}{H}\right), \quad W_{n j}^{(f)} = (-1)^j \exp\left(\frac{2\pi i j \lambda_{L,k+1} x_n}{H}\right), \quad (142)$$

the first of size  $N \times (N/2 + 1)$  for the half-spectrum axis and the second of size  $N \times N$  for the two full axes, both rebuilt whenever  $\lambda_{L,k+1}$  changes. Then (141) is the sequence of contractions

$$\hat{\tilde{u}}_i \xrightarrow{W^{(f)} \text{ on axis 3}} \xrightarrow{W^{(f)} \text{ on axis 2}} \xrightarrow{W^{(1)} \text{ on axis 1}} \operatorname{Re}(\cdot), \quad (143)$$

each of which is a matrix-matrix product once the contracted axis is brought to the front. The cost falls to  $O(d N^{d+1})$ , i.e.  $8 \times 10^8$  multiply-adds at  $N = 128$ , a factor of  $5 \times 10^3$  below the direct sum, and the work is expressed as three dense GEMM calls. The factorization is exact — it is a regrouping of the same finite sum, not an approximation — so the guarantee of §1.3 that the scaling step contributes no spatial discretization error is retained. For  $d = 1$  the sequence (143) collapses to the single contraction with  $W^{(1)}$ , which is (75).

**The Nyquist bin.** For even  $N$  the bins  $j_d = \pm N/2$  of a real field's transform coincide, and the two axis layouts resolve that differently. On the half-spectrum axis the mode is stored once and is given multiplicity  $m_{N/2} = 1$  by (140). On a full axis it is stored in the single bin  $j_d = N/2$  but represents the conjugate pair, so it must be split evenly between  $+N/2$  and  $-N/2$ ; the two halves combine to

$$W_{n, N/2}^{(f)} = (-1)^{N/2} \cos\left(\frac{\pi N \lambda_{L,k+1} x_n}{H}\right), \quad (144)$$

the same cosine that closes (75). With (144) in place the contraction (143) reproduces the direct sum (141) to roundoff for either parity of  $N$ .

**The out-of-box cutoff.** The box indicator  $\chi$  of (28) is a product of one factor per axis, like the kernel, so it separates the same way. It is imposed by zeroing row  $n$  of each interpolation matrix (142) whenever the query of that axis leaves the box,

$$W_{n j_1}^{(1)} \leftarrow 0, \quad W_{n j}^{(f)} \leftarrow 0 \quad \text{whenever } |x_n \lambda_{L,k+1}| > H/2. \quad (145)$$

After the three contractions (143) the point  $\mathbf{x}_n$  then carries the product of the three row factors, which is  $\chi(\mathbf{x}_n \lambda_{L,k+1})$  exactly. The cutoff is thus applied at no cost beyond building the matrices, and the direct sum it is tested against applies  $\chi$  pointwise. For  $\lambda_{L,k+1}$  within a few percent of unity the zeroed rows are the outermost one or two grid points per side. These lie deep in the sponge layer (§5.3), where the field has already been damped, and APPLYBC, when enabled, clamps most of them anyway (§5.11). Zeroing a region in physical space is not a solenoidal operation, so, like the clamp, it reintroduces a divergence proportional to the amplitude it removes.

**Loss of solenoidality and the Leray projection.** The continuous scaling operator preserves continuity: if  $\partial \tilde{u}_j / \partial x_j = 0$ , then  $\partial / \partial x_j [\lambda_U^{-1} \tilde{u}_j(\mathbf{x} \lambda_L)] = \lambda_U^{-1} \lambda_L (\partial \tilde{u}_j / \partial x_j)(\mathbf{x} \lambda_L) = 0$ , the counterpart in physical space of (111). Its discrete evaluation does not preserve it exactly. The dilated field carries the wavenumbers  $\lambda_L \boldsymbol{\kappa}$ , which lie off the lattice of the grid, so sampling (141) at the grid points and transforming back redistributes each mode over neighbouring lattice wavenumbers without regard to  $\kappa_i \hat{u}_i = 0$ ; the out-of-box cutoff (145) further removes amplitude in physical space. Both leave a divergence on the grid. When the option *project* of §2.5 is enabled, SCALE therefore ends with the Leray projection of the scaled field back onto the solenoidal subspace, using the projection tensor (115),

$$\hat{u}_i(\boldsymbol{\kappa}) \leftarrow P_{il}(\boldsymbol{\kappa}) \hat{u}_l(\boldsymbol{\kappa}) = \hat{u}_i - \frac{\kappa_i \kappa_l \hat{u}_l}{\kappa^2}, \quad \boldsymbol{\kappa} \neq \mathbf{0}, \quad (146)$$

followed by the transform back to physical space. The mean mode  $\boldsymbol{\kappa} = \mathbf{0}$  is left unchanged. The projection is orthogonal, so it can only remove energy, and the cost is one forward and one inverse transform per scaling step. Under STRANG it is applied after each of the two scaling half steps, so the field entering EVOLVE is solenoidal as well as the field the step returns.

**Interaction with dealiasing.** The dilation maps each represented wavenumber to  $\lambda_{L,k+1} \boldsymbol{\kappa}$ , so a field that fills the spectrum up to the Nyquist wavenumber would wrap under any  $\lambda_{L,k+1} > 1$ . The 2/3 rule already required by the pseudo-spectral product (102) supplies the margin: the top retained mode sits at  $N/3$  and moves to  $\lambda_{L,k+1} N/3$ , which stays below  $N/2$  for every  $\lambda_{L,k+1} < 3/2$ . Runs in the self-similar frame hold  $\lambda_{L,k+1}$  within a few percent of unity, so the margin is not approached, but it is the condition under which the interpolation stays free of aliasing. Finally, (143) is applied to each velocity component independently and mixes neither components nor the wavenumbers within an axis; at the exceptional value  $\lambda_{L,k+1} = 1$  it reduces to the plain inverse transform, which is taken instead at  $O(N^d \log N)$ .

## 5.11 Boundary treatment

**The clamp.** When enabled, APPLYBC imposes a homogeneous Dirichlet condition on the velocity by clamping the outer layer of grid points of the centred domain (41) after each completed step,

$$\tilde{u}_i(\mathbf{x}_n) \leftarrow 0 \quad \text{whenever } n_d \in \{0, N-1\} \text{ for some direction } d, \quad (147)$$

for every component  $i = 1, 2, 3$ . As in one dimension, on the centred grid  $x_d \in [-H/2, H/2]$  the layers  $n_d = 0$  and  $n_d = N-1$  are distinct grid points rather than periodic images of one another, so both are clamped; the six layers together are the boundary of the box.

Zeroing the faces is not a solenoidal operation, and APPLYBC does not project afterwards: the Leray projection (146), when enabled, is applied only inside SCALE (§5.10). The clamp runs after the last scaling step of the step, so the stored field  $\tilde{u}_{k+1}$  keeps its faces exactly zero and carries the small divergence the clamp injects there. Under STRANG the first scaling half step of the next step projects it away before EVOLVE; under LIETROTTER EVOLVE starts from it. The divergence stays small only while the field is negligible at the faces, which the sponge forcing (§5.3) is there to ensure.

## 6 Navier–Stokes results

### 6.1 Kinetic energy budget in the self-similar frame

We work with the full velocity  $u_i^*$  and the ensemble average  $\overline{\cdot}$  of §5.7.1, and assume that the ensemble-mean velocity vanishes,  $\overline{u_i^*} = 0$ . Then  $u_i^*$  is the fluctuating velocity, the mean-advection and production terms of the turbulent kinetic energy equation [2] drop out, and the mean of the kinetic energy  $\overline{k^*} = \frac{1}{2} \overline{u_i^* u_i^*}$ , obeys

$$\frac{\partial \overline{k^*}}{\partial t^*} = \overline{T^*} - \overline{\epsilon^*}, \quad \overline{T^*} = -\frac{\partial}{\partial x_j^*} \left( \frac{1}{\rho} \overline{p^* u_j^*} + \frac{1}{2} \overline{u_i^* u_i^* u_j^*} - 2\nu \overline{u_i^* S_{ij}^*} \right), \quad (148)$$

where  $\overline{T^*}$  is the transport,  $S_{ij}^*$  is the strain rate of  $u_i^*$ , and  $\overline{\epsilon^*} = 2\nu \overline{S_{ij}^* S_{ij}^*}$  is the mean of the dissipation rate of §5.7.2.

We nondimensionalize with (2), the pressure scale (81), and

$$\bar{k}^* = U^2 \bar{k}, \quad S_{ij}^* = \frac{U}{L} S_{ij}, \quad \bar{\epsilon}^* = \frac{U^3}{L} \bar{\epsilon}, \quad (149)$$

as in (124) and §5.7.2. Since  $U$  and  $L$  are the same for every member of the ensemble (assumption 1 of §5.7.1), they pass through the averages, e.g.  $\overline{u_i^* u_i^* u_j^*} = U^3 \overline{u_i u_i u_j}$ . The pressure and triple-correlation fluxes in  $\bar{T}^*$  therefore carry  $U^3$ , the viscous flux carries  $\nu U^2/L$ , and each derivative contributes  $\partial/\partial x_j^* = L^{-1} \partial/\partial x_j$ , so that

$$\bar{T}^* = \frac{U^3}{L} \bar{T}, \quad \bar{T} = -\frac{\partial}{\partial x_j} \left( \overline{p u_j} + \frac{1}{2} \overline{u_i u_i u_j} - \frac{2}{Re} \overline{u_i S_{ij}} \right), \quad (150)$$

with  $Re$  the Reynolds number (87). In the limit  $1/Re = 0$  in which we run, the viscous term drops out of  $\bar{T}$ , while  $\bar{\epsilon}$  stays finite and is measured through the subgrid transfer by (122).

The left-hand side of (148) follows from the chain rule exactly as in (6), with the amplitude  $U^2$  in place of  $U$ , so that  $\sigma_U$  is replaced by  $(U^2)' / U^2 = 2\sigma_U$ ,

$$\frac{\partial \bar{k}^*}{\partial t^*} = \frac{U^2}{\tau} \left( \frac{\partial \bar{k}}{\partial t} + 2\sigma_U \bar{k} - \sigma_L x_j \frac{\partial \bar{k}}{\partial x_j} \right). \quad (151)$$

With the clock (85),  $U^2/\tau = U^3/L$ , so the prefactors of (151) and (150) cancel and (148) becomes

$$\frac{\partial \bar{k}}{\partial t} = \sigma_L x_j \frac{\partial \bar{k}}{\partial x_j} - 2\sigma_U \bar{k} + \bar{T} - \bar{\epsilon}. \quad (152)$$

## 6.2 Stationary budget

In the self-similar frame the flow is statistically stationary, so  $\bar{k}$ ,  $\bar{T}$ , and  $\bar{\epsilon}$  are independent of  $t$  and (152) reduces to

$$0 = \sigma_L x_j \frac{\partial \bar{k}}{\partial x_j} - 2\sigma_U \bar{k} + \bar{T} - \bar{\epsilon}. \quad (153)$$

Since  $\bar{T} - \bar{\epsilon}$  is independent of  $t$ , so are the scaling terms. Under the power-law ansatz the rates are  $\sigma_X = \beta_X \tau / (t^* + \alpha_X)$  by (51). Taking  $\alpha_U = \alpha_L = \alpha$ , the clock (85) is  $\tau = (t^* + \alpha)^{\beta_L - \beta_U}$ , and

$$\sigma_L x_j \frac{\partial \bar{k}}{\partial x_j} - 2\sigma_U \bar{k} = (t^* + \alpha)^{\beta_L - \beta_U - 1} \left( \beta_L x_j \frac{\partial \bar{k}}{\partial x_j} - 2\beta_U \bar{k} \right). \quad (154)$$

Either the bracket vanishes identically, which by (153) would require  $\bar{T} = \bar{\epsilon}$  everywhere, or the prefactor is constant,  $\beta_L - \beta_U = 1$ ; the numerical experiments confirm the latter. The clock is then  $\tau = t^* + \alpha$ , the rates  $\sigma_X = \beta_X$  are constant, the self-similar time is  $t = \ln(t^* + \alpha)$  up to a constant, and (153) holds with constant coefficients,

$$0 = \beta_L x_j \frac{\partial \bar{k}}{\partial x_j} - 2\beta_U \bar{k} + \bar{T} - \bar{\epsilon}. \quad (155)$$

In physical variables,  $\bar{T}^* - \bar{\epsilon}^* = (U^3/L)(\bar{T} - \bar{\epsilon})$  by (149) and (150), and  $U^3/L = U^2/\tau = (t^* + \alpha)^{2\beta_U - 1}$ , so (148) and (155) give

$$\frac{\partial \bar{k}^*}{\partial t^*} = (t^* + \alpha)^{2\beta_U - 1} \left( 2\beta_U \bar{k} - \beta_L x_j \frac{\partial \bar{k}}{\partial x_j} \right). \quad (156)$$

At the origin  $x_j \partial \bar{k} / \partial x_j = 0$ , and (155) gives the energy there directly,

$$\bar{k}(0) = \frac{\bar{T}(0) - \bar{\epsilon}(0)}{2\beta_U}. \quad (157)$$

Then (156) integrates to  $\bar{k}^*(0, t^*) = (t^* + \alpha)^{2\beta_U} \bar{k}(0)$ , the scaling (??).

Away from the origin we assume that the ensemble statistics are spherically symmetric about it, so that  $\bar{k}$ ,  $\bar{T}$ , and  $\bar{\epsilon}$  depend only on  $r = \sqrt{x_i x_i}$ . Then  $x_j \partial \bar{k} / \partial x_j = r d\bar{k} / dr$ , and (155) becomes the ordinary differential equation

$$\beta_L r \frac{d\bar{k}}{dr} - 2\beta_U \bar{k} = \bar{\epsilon} - \bar{T}. \quad (158)$$

With the integrating factor  $r^{-2\beta_U/\beta_L}$  it becomes

$$\frac{d}{dr} \left( r^{-2\beta_U/\beta_L} \bar{k} \right) = \frac{1}{\beta_L} r^{-2\beta_U/\beta_L-1} (\bar{\epsilon} - \bar{T}). \quad (159)$$

The homogeneous solution is  $\bar{k} = C r^{2\beta_U/\beta_L}$ . For a flow whose length scale grows and whose velocity scale decays,  $\beta_L > 0 > \beta_U$ , and this is singular at the origin. Requiring  $\bar{k}$  to be finite there sets  $C = 0$ , and integrating (159) from 0 to  $r$  gives

$$\bar{k}(r) = \frac{1}{\beta_L} \int_0^r \left( \frac{s}{r} \right)^{-2\beta_U/\beta_L} (\bar{\epsilon}(s) - \bar{T}(s)) \frac{ds}{s}. \quad (160)$$

Finally, with  $\bar{k}^* = U^2 \bar{k}$  and  $r = r^*/L$ , the physical energy is

$$\bar{k}^*(r^*, t^*) = (t^* + \alpha)^{2\beta_U} \bar{k} \left( \frac{r^*}{(t^* + \alpha)^{\beta_L}} \right). \quad (161)$$

## A The RK4 physics step

This appendix sets out the scheme used for the Navier–Stokes physics operator  $\Phi^{(P)}$  of §5.6: one step of the classical fourth-order Runge–Kutta method. Throughout we abbreviate the step as  $h \equiv \Delta t$ , reserving  $\Delta$  for the filter width of the subgrid stress model of §5.2.

### A.1 The scheme

Written out mode by mode, the physics flow (117) is the system of ordinary differential equations

$$\frac{d\hat{u}_i}{dt} = R_i(\hat{u}), \quad R_i(\hat{u}) = -P_{il}(\boldsymbol{\kappa}) \left( \hat{N}_l - \hat{G}_l + \hat{F}_l \right), \quad (162)$$

the bracketed physical terms of (116). RK4 advances it through four evaluations of  $R$ ,

$$\begin{aligned} \hat{k}_i^{(1)} &= R_i(\hat{u}^{(n)}), & \hat{k}_i^{(2)} &= R_i(\hat{u}^{(n)} + \frac{h}{2} \hat{k}^{(1)}), \\ \hat{k}_i^{(3)} &= R_i(\hat{u}^{(n)} + \frac{h}{2} \hat{k}^{(2)}), & \hat{k}_i^{(4)} &= R_i(\hat{u}^{(n)} + h \hat{k}^{(3)}), \end{aligned} \quad (163)$$

combined with Simpson weights,

$$\hat{u}_i^{(n+1)} = \hat{u}_i^{(n)} + \frac{h}{6} \left( \hat{k}_i^{(1)} + 2\hat{k}_i^{(2)} + 2\hat{k}_i^{(3)} + \hat{k}_i^{(4)} \right). \quad (164)$$

The convective part of each evaluation is computed pseudo-spectrally in rotational form,  $\tilde{u}_j \partial_j \tilde{u}_i = -(\tilde{\mathbf{u}} \times \tilde{\boldsymbol{\omega}})_i + \partial_i(\tilde{u}_k \tilde{u}_k/2)$ , whose gradient part  $P_{il}$  annihilates; this costs six inverse and three forward transforms. Under the Smagorinsky model the subgrid stress term (105) costs six further inverse transforms, one per independent component of  $\tilde{S}_{ij}$ , and six forward transforms of  $T_{ij}$ ; under the self-similar constant eddy viscosity it is the diagonal multiplication (106) and costs none. The sponge term (107) reuses the physical-space velocity already formed for the convective term and multiplies it by the tabulated  $W$ , so it costs only the three forward transforms of  $W \tilde{u}_i$ . An evaluation thus costs 24 transforms under the Smagorinsky model and 12 under the constant eddy viscosity, and a step 96 or 48 in all. The scheme has no coefficients beyond the four weights, so nothing is precomputed or cached.

**Order.** The scheme is fourth order in  $h$ , so it satisfies the second-order requirement §2.5 places on EVOLVE for the Strang composition with a margin.

## A.2 Stability

Applied to the linear model  $d\hat{u}/dt = \lambda\hat{u}$ , (163)–(164) multiply  $\hat{u}$  by the amplification factor

$$A(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}, \quad z = \lambda h, \quad (165)$$

the degree-four Taylor polynomial of  $e^z$ , and the step is stable while  $|A(z)| \leq 1$ .

**The advective limit.** The convective term is, locally, a transport at speed  $|\tilde{\mathbf{u}}|$ , whose eigenvalues on the retained band lie on the imaginary axis with  $|\lambda| \leq \max |\tilde{\mathbf{u}}| \kappa_{\max}$ , where  $\kappa_{\max} = 2\pi(N/3)/H$  is the top mode retained by the 2/3 rule. Along the imaginary axis  $|A(z)| \leq 1$  holds for  $|z| \leq 2\sqrt{2}$ , so

$$h \max |\tilde{\mathbf{u}}| \kappa_{\max} \leq 2\sqrt{2}. \quad (166)$$

The self-similar frame pins  $\max |\tilde{\mathbf{u}}| = O(1)$  on average, so the limit does not drift over a run as it would in a decaying simulation in physical variables. It does not pin it pointwise in time, however: the field can momentarily concentrate into a high-amplitude structure whose  $\max |\tilde{\mathbf{u}}|$  carries a step that is stable in the settled regime past (166).

**The sponge limit.** The sponge forcing (97) is, pointwise, a linear decay at rate  $W(r) \in [0, W_{\max}]$ , whose eigenvalues lie on the negative real axis with  $|\lambda| \leq W_{\max}$ . Along the negative real axis  $|A(z)| \leq 1$  holds for  $-2.785 \lesssim z \leq 0$ , so

$$h W_{\max} \leq 2.785. \quad (167)$$

Unlike (166), this limit does not depend on the state: it is fixed by the run parameter  $W_{\max}$ . It is not enforced by MAXSTABLEDT, but assumed. At the default  $W_{\max} = 10$  and a CFL-limited step of  $h \approx 0.05$ ,  $h W_{\max} \approx 0.5$ , well inside both (167) and the range  $h W_{\max} \lesssim 1$  in which the step also follows the exponential decay  $e^{-Wh}$  accurately. A run with a larger  $W_{\max}$  or a longer step must check (167) by hand.

**The diffusive limit.** Under the self-similar constant eddy viscosity, the subgrid stress term (106) is marched explicitly with the rest of  $R_i$ . It is a linear decay at rate  $C_\nu \Delta^2 \kappa^2$ , whose eigenvalues lie on the negative real axis with  $|\lambda| \leq C_\nu \Delta^2 \kappa_{\max}^2$ , so by the same bound as (167)

$$h C_\nu \Delta^2 \kappa_{\max}^2 \leq 2.785. \quad (168)$$

With  $\Delta = \Delta x = H/N$  and  $\kappa_{\max} = 2\pi(N/3)/H$ , the product  $\Delta \kappa_{\max} = 2\pi/3$  is independent of the grid, and the limit becomes

$$h C_\nu \leq \frac{9 \times 2.785}{4\pi^2} \approx 0.635. \quad (169)$$

Like the sponge limit it does not depend on the state, since in the self-similar frame  $\nu_t = C_\nu \Delta^2$  is fixed for the run, and because the eddy viscosity scales with  $\Delta^2$  it does not tighten under grid refinement either. It is not enforced by MAXSTABLEDT, but assumed, and can be guaranteed by choosing  $\Delta t_{\max} \leq 0.635/C_\nu$ . At a CFL-limited step of  $h \approx 0.05$  it holds for  $C_\nu \lesssim 12.7$ .

**The CFL time step.** The advective limit is therefore enforced, by the step rule (44) of §2.3. Its CFL term, MAXSTABLEDT, is the step that puts (166) at the fraction  $C$  of its limit,

$$\Delta t_{\text{CFL}}(\tilde{u}; C) = C \frac{2\sqrt{2}}{\max |\tilde{\mathbf{u}}| \kappa_{\max}}, \quad (170)$$

and the realized CFL (45) is  $\Delta t_k \max |\tilde{\mathbf{u}}_{k+1}| \kappa_{\max} / 2\sqrt{2}$ .

**A resolution margin.** A structure that concentrates below the grid scale also raises  $\max |\tilde{\mathbf{u}}|$ , and so looks like a CFL problem; but a smaller step only delays it. To tell the two apart the run also records the fraction of the dealiased energy  $\frac{1}{2} \sum |\hat{u}|^2$  carried by  $|\kappa| > \frac{2}{3} \kappa_{\max}$ . It stays small while the field is resolved and rises as energy piles up against the 2/3 cutoff. A realized CFL excursion with a small tail fraction is a time-step problem; one preceded by a rising tail fraction is a resolution problem.

### A.3 Solenoidality

The step contributes nothing of its own to the continuity residual. Every contribution to  $R_i$  — convective, subgrid, and sponge alike — has had  $P_{il}$  applied, and  $\kappa_i P_{il} = \kappa_l - \kappa_l \kappa^2 / \kappa^2 = 0$ , so contracting (162) with  $\kappa_i$  gives  $\kappa_i R_i(\hat{u}) = 0$ . Each stage of (163) is then divergence free, and the divergence of each mode is carried through the step unchanged,

$$\kappa_i \hat{u}_i^{(n+1)} = \kappa_i \hat{u}_i^{(n)}. \quad (171)$$

Whatever divergence enters the field is therefore neither amplified nor damped by the step, and if (109) holds on input it holds on output. The residual the run log records is therefore entirely the boundary clamp's, when the clamp is enabled (§5.11), and not the time stepper's.

### A.4 Dealiasing

The 2/3 mask of §5.4 is applied to the incoming field before the first stage. No stage then leaves the retained band: the convective, subgrid stress, and sponge terms are masked after their forward transforms, the projection  $P_{il}$  is diagonal in  $\kappa$ , and each stage argument in (163) is a linear combination of band-limited fields. The mask is applied once more to  $\hat{u}^{(n+1)}$ , which changes nothing in exact arithmetic but makes explicit that no content reaches the band the next scaling step will stretch.

## B Statistical error of time averages

This appendix sets out how the statistical error of a time-averaged quantity such as (121) is estimated, following [3]. Let  $q(t)$  be a statistically stationary signal, here the nondimensional subgrid transfer at the origin,  $q(t) = \Pi(0, t)$  of (123), or the resolved kinetic energy there,  $q(t) = \tilde{k}(0, t)$  of (124), both in the self-similar frame, with mean  $\langle q \rangle$ , fluctuation  $q' = q - \langle q \rangle$ , and root-mean-square fluctuation  $q'_{\text{rms}}$ . Because the clock (85) is the advective time scale  $L/U$ , the self-similar time  $t$  is already measured in eddy turnover times.

### B.1 Batch means

Successive values of  $q$  are correlated, so the samples of a run cannot be treated as independent. We instead use the method of batch means:

1. Discard the initial transient, during which the flow approaches the self-similar state, and let  $T$  be the length of the remaining record.
2. Split the record into  $M = T/T_w$  non-overlapping windows of length  $T_w$ , and compute the window means

$$\langle q \rangle_i = \frac{1}{T_w} \int_{t_i}^{t_i + T_w} q(t) dt, \quad i = 1, \dots, M. \quad (172)$$

The samples  $q_k = q(t_k)$  are recorded once per step, and under the CFL time step of §2.3 the  $t_k$  are unequally spaced. Each integral (172) is therefore evaluated as the exact integral of the piecewise-linear interpolant of the samples: the trapezoid rule on  $(t_k, q_k)$ , with  $q$  interpolated linearly to window edges that fall between samples. The windows then tile  $[t_0, t_0 + MT_w]$  exactly, and the remainder of the record shorter than  $T_w$  is dropped. The plain mean of the samples in a window would not do: it weights each sample by one rather than by its  $\Delta t_k$ , and since  $\Delta t_k \propto 1/\max |u|$  is correlated with the flow, the resulting mean is biased.

3. Estimate the mean and its standard error (SEM) as

$$q_{\text{est}} = \frac{1}{M} \sum_{i=1}^M \langle q \rangle_i, \quad \text{SEM} \simeq \frac{\text{STD}(\langle q \rangle_i)}{\sqrt{M}}. \quad (173)$$

4. Report the 95% confidence interval  $q_{\text{est}} \pm 1.96 \text{ SEM}$ .

The confidence interval assumes a Gaussian error, which the central limit theorem supplies when the windows are uncorrelated. Equation (173) holds only if

1. the windows are uncorrelated,  $T_w \gg \tau_{\text{corr}}$ , where  $\tau_{\text{corr}}$  is the integral time scale of  $q$ ; and
2. there are enough windows,  $M \gg 1$ .

For a fixed  $T$  the two conditions pull in opposite directions, so  $T_w$  has to be chosen with care.

## B.2 Choosing the window length

The window length is chosen by plotting  $\text{SEM}\sqrt{T}$  against  $T_w$ , both in units of the eddy turnover time. The curve has three regimes:

- *Small  $T_w$ .* Adjacent windows are correlated, so  $M$  overcounts the independent samples and  $\text{SEM} \approx q'_{\text{rms}} \sqrt{T_w/T}$ , a slope of 1/2 on log axes. This underestimates the true error.
- *Plateau.* Both conditions hold and  $\text{SEM} \approx q'_{\text{rms}} \sqrt{\tau_{\text{corr}}/T}$ , independent of  $T_w$ . This is the true SEM.
- *Large  $T_w$ .*  $M$  is too small and the estimate is noisy and unreliable.

The small- $T_w$  asymptote and the plateau intersect at  $T_w = \tau_{\text{corr}}$ , where

$$\tau_{\text{corr}} = \frac{1}{q'^2_{\text{rms}}} \int_{-\infty}^{\infty} C(\tau) d\tau, \quad C(\tau) = \langle q'(t) q'(t + \tau) \rangle, \quad (174)$$

so the plot also yields the integral time scale. A clear plateau requires a long record: [3] found one only for  $T \gtrsim 10^4$  eddy turnover times, approximately  $1.6 \times 10^4$  for homogeneous isotropic turbulence and  $1.4 \times 10^4$  for channel flow at  $Re_\tau = 180$ . A dissipation-type quantity, being a second-order statistic of the velocity gradients, has  $\tau_{\text{corr}}$  of roughly five eddy turnover times in homogeneous isotropic turbulence.

## References

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